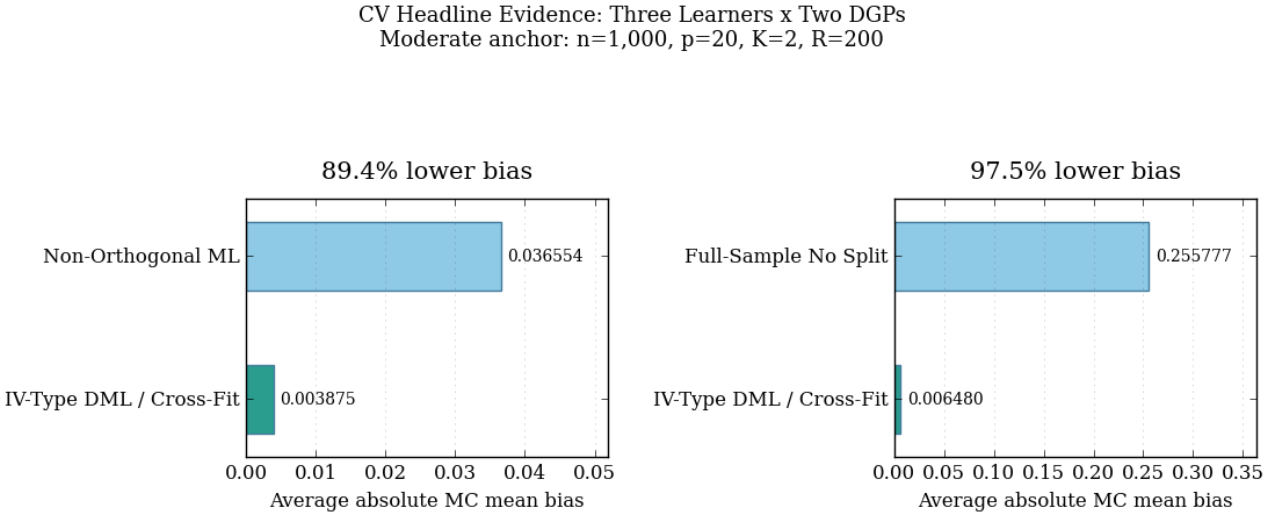
# Double Machine Learning: CV Evidence

A focused companion to the research-CV project description. This document maps each claim to saved Monte Carlo evidence; it is not a replacement for the CV and does not add new model fits.

## 1. Main comparison and metric

Three nuisance learner families (Lasso, Random Forest and XGBoost) are evaluated in linear and nonlinear moderate DGPs. Both headline comparisons use n=1,000, p=20, true effect 0.5, K=2, the base learner preset and 200 replications per setting.




Headline metric

| Comparison | Before | After | Reduction |
|---|---|---|---|
| Non-orthogonal ML vs. DML | 0.0365543 | 0.0038746 | 89.4% |
| Full no-split vs. cross-fitting | 0.2557766 | 0.0064802 | 97.5% |

For each learner-DGP setting, bias is the mean of theta_hat minus 0.5 over 200 replications. The headline averages the absolute values of those six signed means. It is NOT MAE, RMSE, the average of six percentage reductions, or a result pooled over the sensitivity grid.

These n=1,000 moderate results are separate from the newly completed n=500 baseline bridge and the reused n=300 baseline report. Raw-error and studentized-error plots below are explicitly labelled.

# 2. Orthogonalization: Linear DGP

Table 1: Non-Orthogonal ML vs. DML (Linear Moderate DGP, n=1,000, R=200)

| Learner | Non-Orthogonal ML | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | 0.0540 | 0.0276 | 0.0214 | 0.0128 | 0.0315 | 0.0312 |
| XGBoost | 0.0464 | 0.0292 | 0.0216 | 0.0050 | 0.0326 | 0.0330 |
| Lasso | 0.0125 | 0.0313 | 0.0211 | 0.0008 | 0.0318 | 0.0321 |

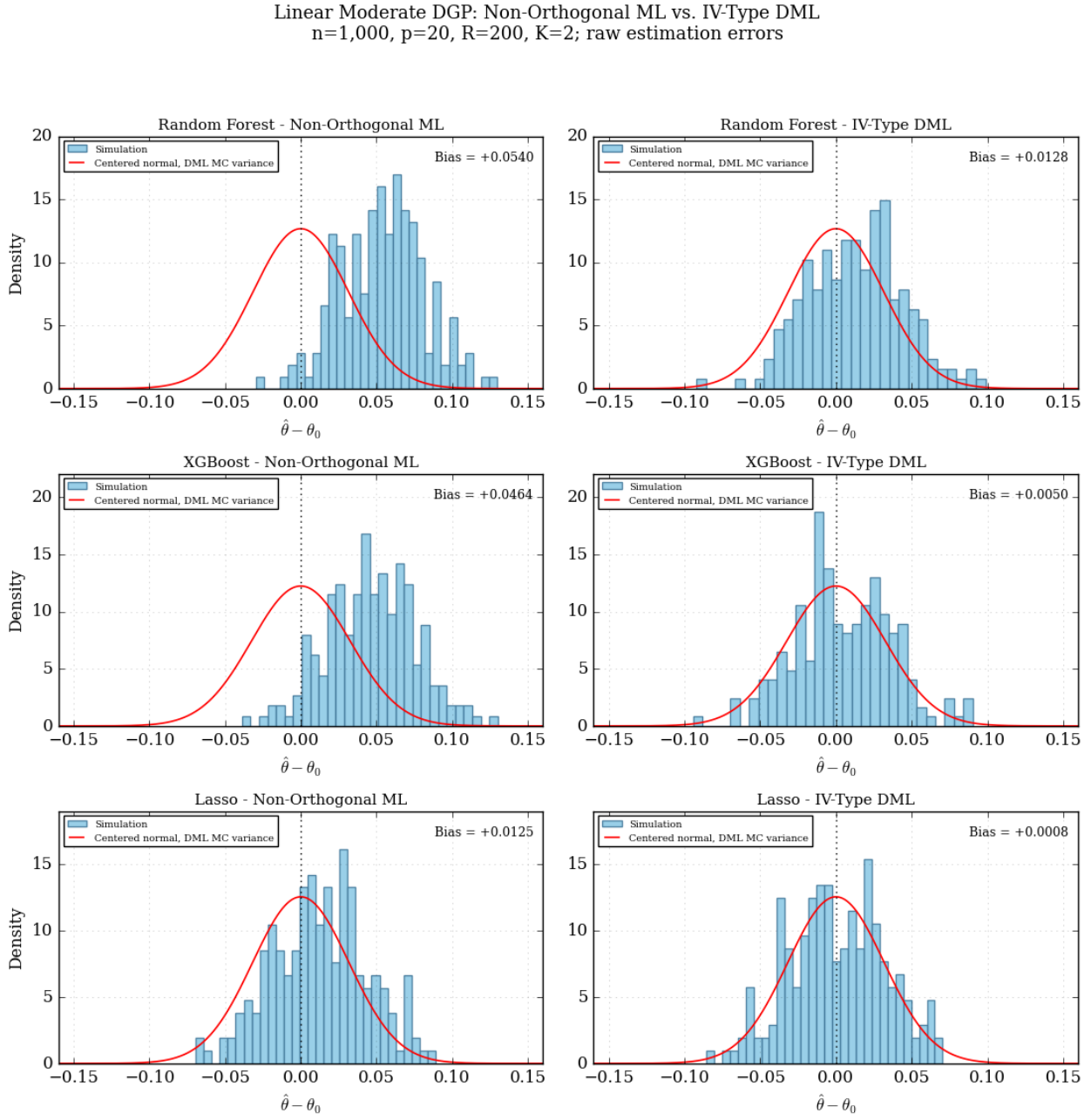



Figure 1. Non-orthogonal ML and IV-type DML use the same out-of-sample structural-nuisance predictions. The estimator score is the changed component. Raw errors are shown; axes match within each learner pair and every draw is retained.

## 2. Orthogonalization: Nonlinear DGP

Table 2: Non-Orthogonal ML vs. DML (Nonlinear Moderate DGP, n=1,000, R=200)

| Learner | Non-Orthogonal ML | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | 0.0405 | 0.0286 | 0.0229 | -0.0014 | 0.0331 | 0.0327 |
| XGBoost | 0.0346 | 0.0301 | 0.0230 | -0.0022 | 0.0345 | 0.0332 |
| Lasso | 0.0314 | 0.0318 | 0.0227 | -0.0012 | 0.0335 | 0.0322 |

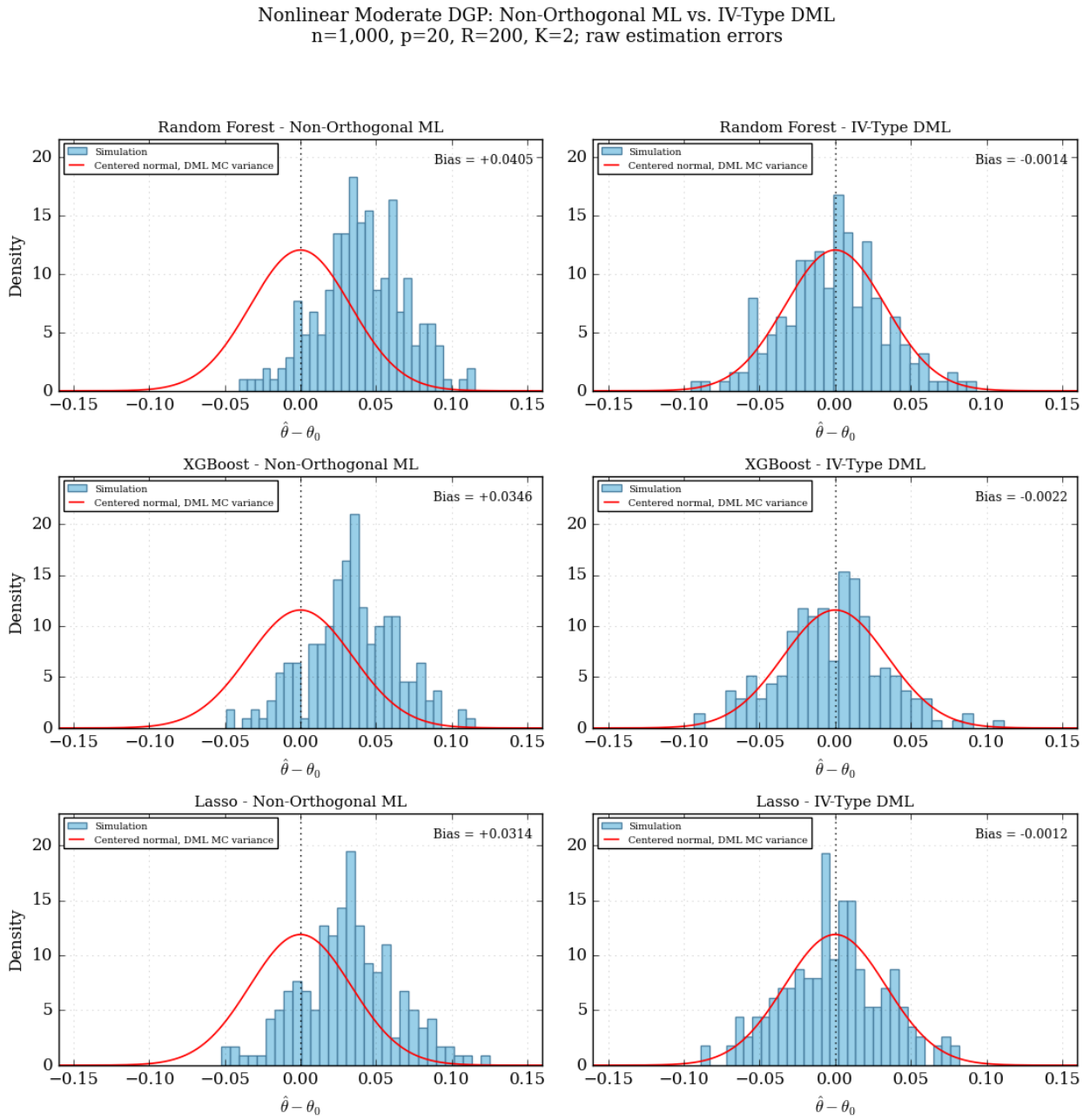



Figure 2. Non-orthogonal ML and IV-type DML use the same out-of-sample structural-nuisance predictions. The estimator score is the changed component. Raw errors are shown; axes match within each learner pair and every draw is retained.

# 3. Learner choice and finite-sample spread

Table 3: Learner Sensitivity (Linear Moderate DGP, n=1,000, R=200)

| Learner | Mean estimate | Bias | MC SD | 95% MC spread | Coverage |
|---|---|---|---|---|---|
| Random Forest | 0.5128 | 0.0128 | 0.0315 | [0.4511, 0.5745] | 93.5% |
| XGBoost | 0.5050 | 0.0050 | 0.0326 | [0.4411, 0.5689] | 94.5% |
| Lasso | 0.5008 | 0.0008 | 0.0318 | [0.4385, 0.5631] | 95.0% |

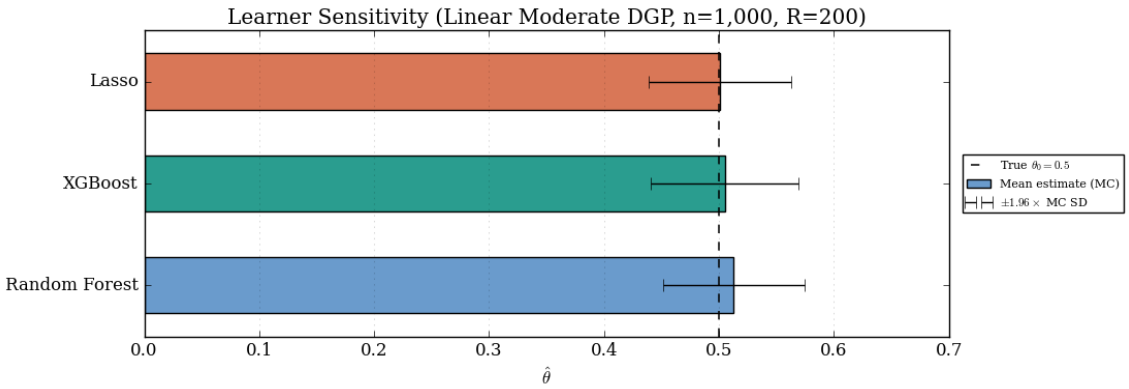
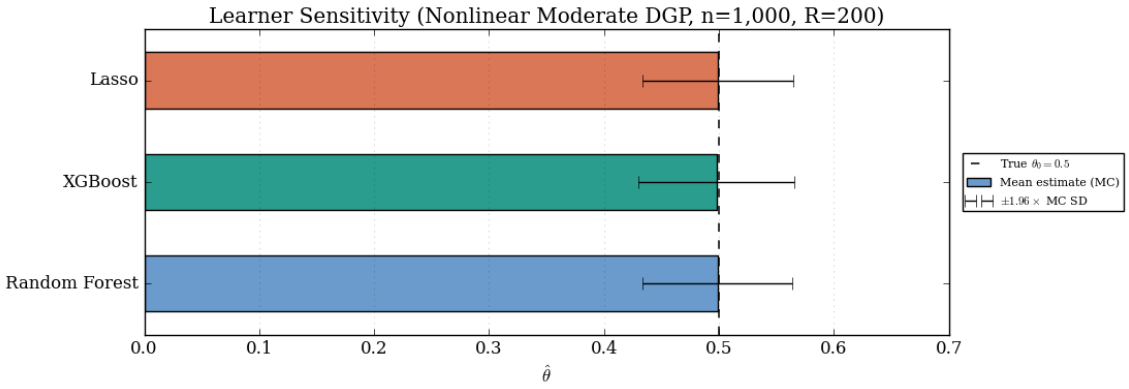



Table 4: Learner Sensitivity (Nonlinear Moderate DGP, n=1,000, R=200)

| Learner | Mean estimate | Bias | MC SD | 95% MC spread | Coverage |
|---|---|---|---|---|---|
| Random Forest | 0.4986 | -0.0014 | 0.0331 | [0.4337, 0.5636] | 94.5% |
| XGBoost | 0.4978 | -0.0022 | 0.0345 | [0.4303, 0.5653] | 94.5% |
| Lasso | 0.4988 | -0.0012 | 0.0335 | [0.4331, 0.5646] | 93.5% |




Error bars are mean estimate +/-1.96 Monte Carlo SD: an approximate sampling spread, NOT a confidence interval for the Monte Carlo mean. Learner-dependent residual bias remains; these results do not establish exact finite-sample unbiasedness.

# 4. Cross-fitting: Linear DGP

Table 5: Full-Sample No Split vs. Cross-Fitting (Linear Moderate DGP, n=1,000, R=200)

| Learner | Full-Sample No Split | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | -0.2704 | 0.0159 | 0.0168 | 0.0115 | 0.0385 | 0.0310 |
| XGBoost | -0.4625 | 0.0054 | 0.0217 | 0.0019 | 0.0411 | 0.0358 |
| Lasso | -0.0130 | 0.0335 | 0.0309 | 0.0050 | 0.0374 | 0.0332 |

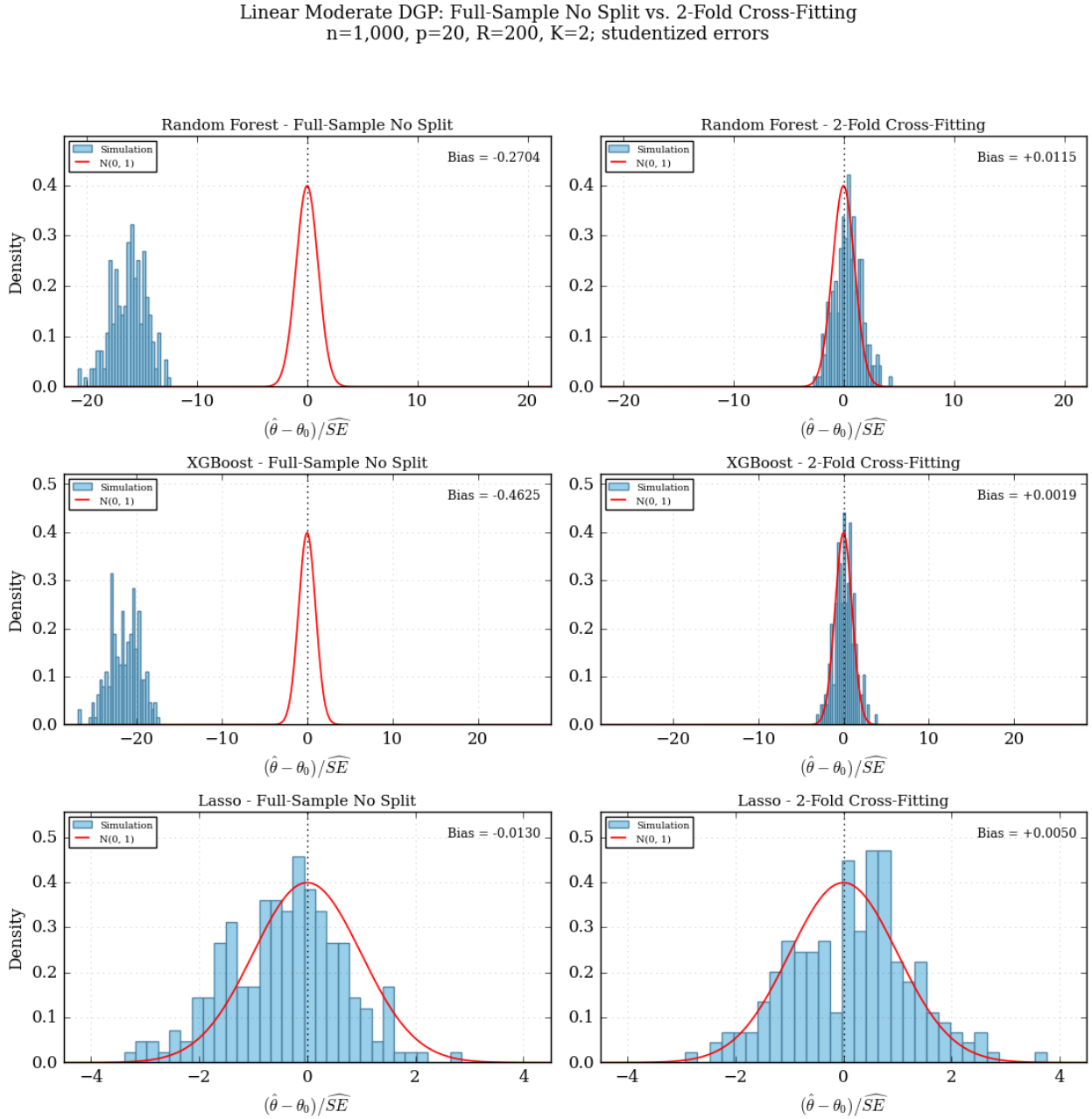



Figure 5. Studentized errors (theta_hat - 0.5)/SE_hat. Full no-split trains on all 1,000 observations; each cross-fit nuisance model trains on 500. Same learner presets and data are used, with no comparator-only stress. Large no-split t-errors are not clipped. Raw-error companion plots are in the appendix.

# 4. Cross-fitting: Nonlinear DGP

Table 6: Full-Sample No Split vs. Cross-Fitting (Nonlinear Moderate DGP, n=1,000, R=200)

| Learner | Full-Sample No Split | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | -0.2941 | 0.0134 | 0.0180 | -0.0073 | 0.0324 | 0.0337 |
| XGBoost | -0.4768 | 0.0036 | 0.0200 | -0.0121 | 0.0346 | 0.0362 |
| Lasso | -0.0180 | 0.0306 | 0.0313 | -0.0010 | 0.0332 | 0.0336 |

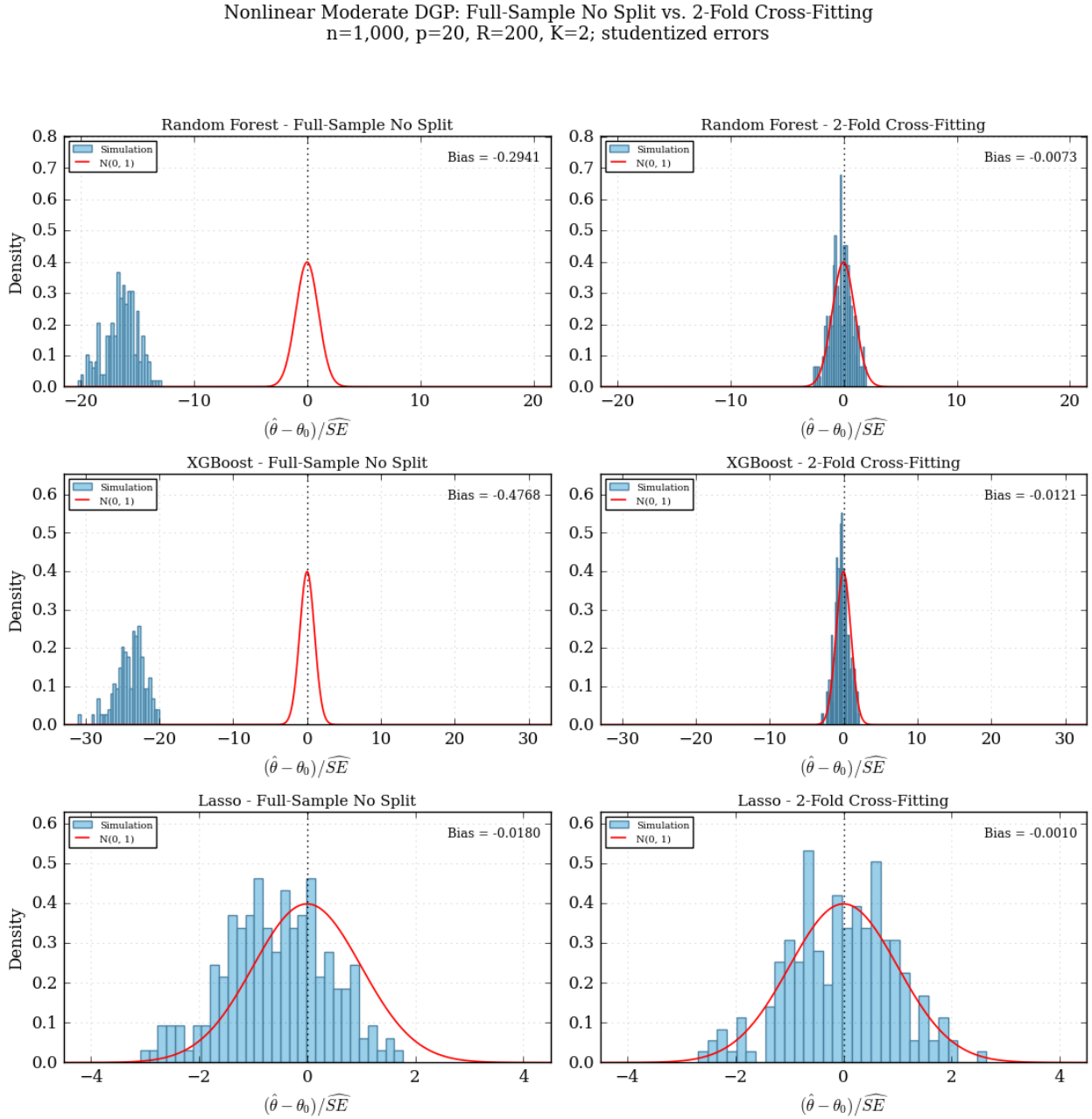



Figure 6. Studentized errors (theta_hat - 0.5)/SE_hat. Full no-split trains on all 1,000 observations; each cross-fit nuisance model trains on 500. Same learner presets and data are used, with no comparator-only stress. Large no-split t-errors are not clipped. Raw-error companion plots are in the appendix.

## 5. Equal-training-size reuse control

To separate sample reuse from training size, equal-training reuse and cross-fitting share exactly the same fitted nuisance-model bank and train on 500 observations per model. Their predictions differ in whether the evaluated outcome was in the training set. The conventional full-sample comparison is reported separately.

Table 7: Equal-Training Reuse vs. Cross-Fitting (Linear Moderate DGP, n=1,000, R=200)

| Learner | Equal-Training Reuse | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | -0.2576 | 0.0166 | 0.0170 | 0.0115 | 0.0385 | 0.0310 |
| XGBoost | -0.4672 | 0.0049 | 0.0233 | 0.0019 | 0.0411 | 0.0358 |
| Lasso | -0.0312 | 0.0333 | 0.0306 | 0.0050 | 0.0374 | 0.0332 |

Table 8: Equal-Training Reuse vs. Cross-Fitting (Nonlinear Moderate DGP, n=1,000, R=200)

| Learner | Equal-Training Reuse | | | IV-type DML | | |
|---|---|---|---|---|---|---|
| | Bias | MC SD | Mean SE | Bias | MC SD | Mean SE |
| Random Forest | -0.2855 | 0.0143 | 0.0184 | -0.0073 | 0.0324 | 0.0337 |
| XGBoost | -0.4794 | 0.0030 | 0.0212 | -0.0121 | 0.0346 | 0.0362 |
| Lasso | -0.0353 | 0.0308 | 0.0309 | -0.0010 | 0.0332 | 0.0336 |

The shared-bank control supports the reuse mechanism but is not relabelled as the full-sample no-split comparator. Its numerical reductions are not substituted into the CV headline.

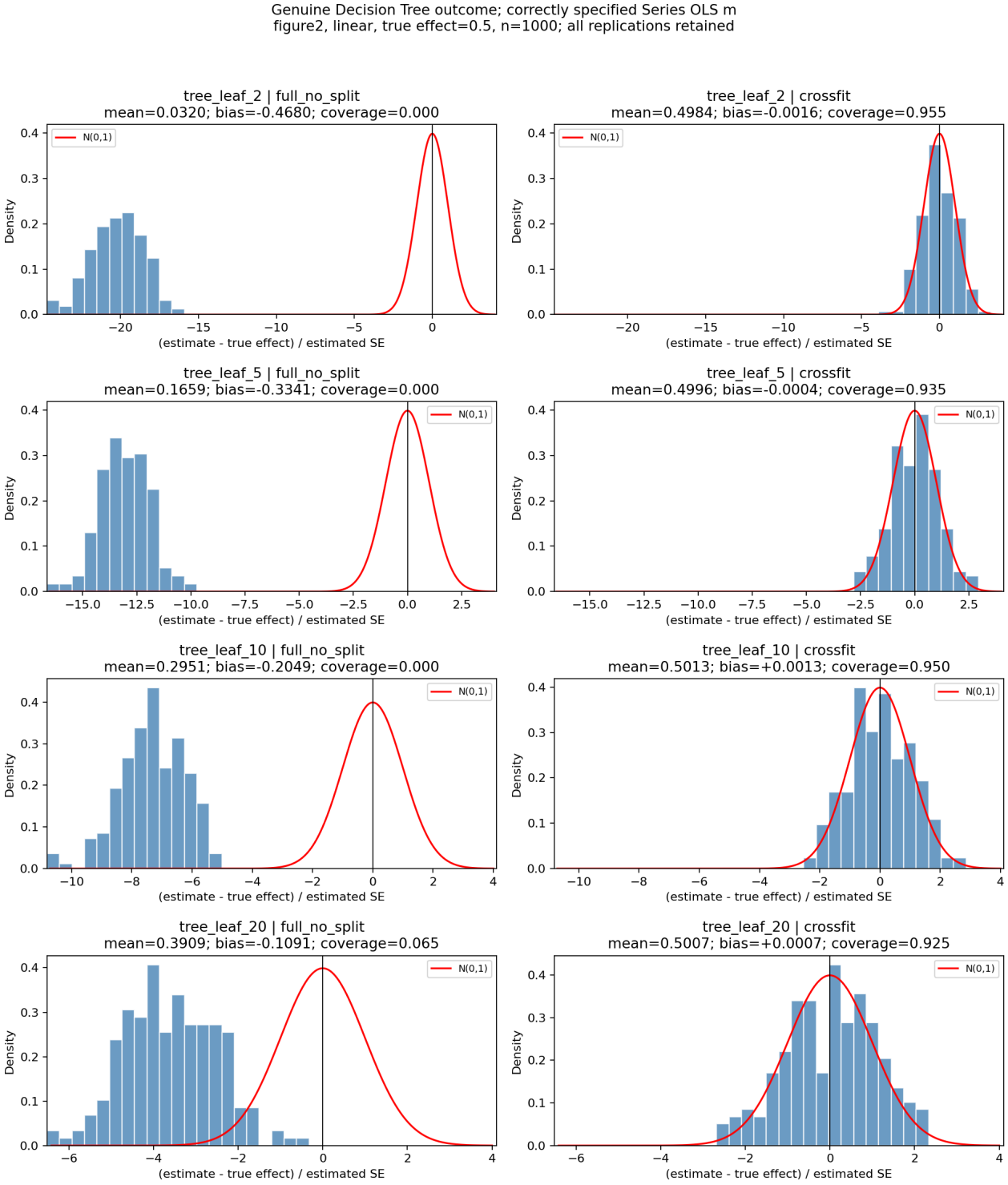
## 6. Controlled Series OLS / decision-tree evidence

A separate controlled study fixes the treatment regression to Series OLS with features [1, X1, X3, sigmoid(X1), sigmoid(X3)] and varies decision-tree outcome leaf size (2, 5, 10, 20). Its n=1,000/R=200 Figure 2 outcomes are not pooled with the Lasso/RF/XGBoost headline.

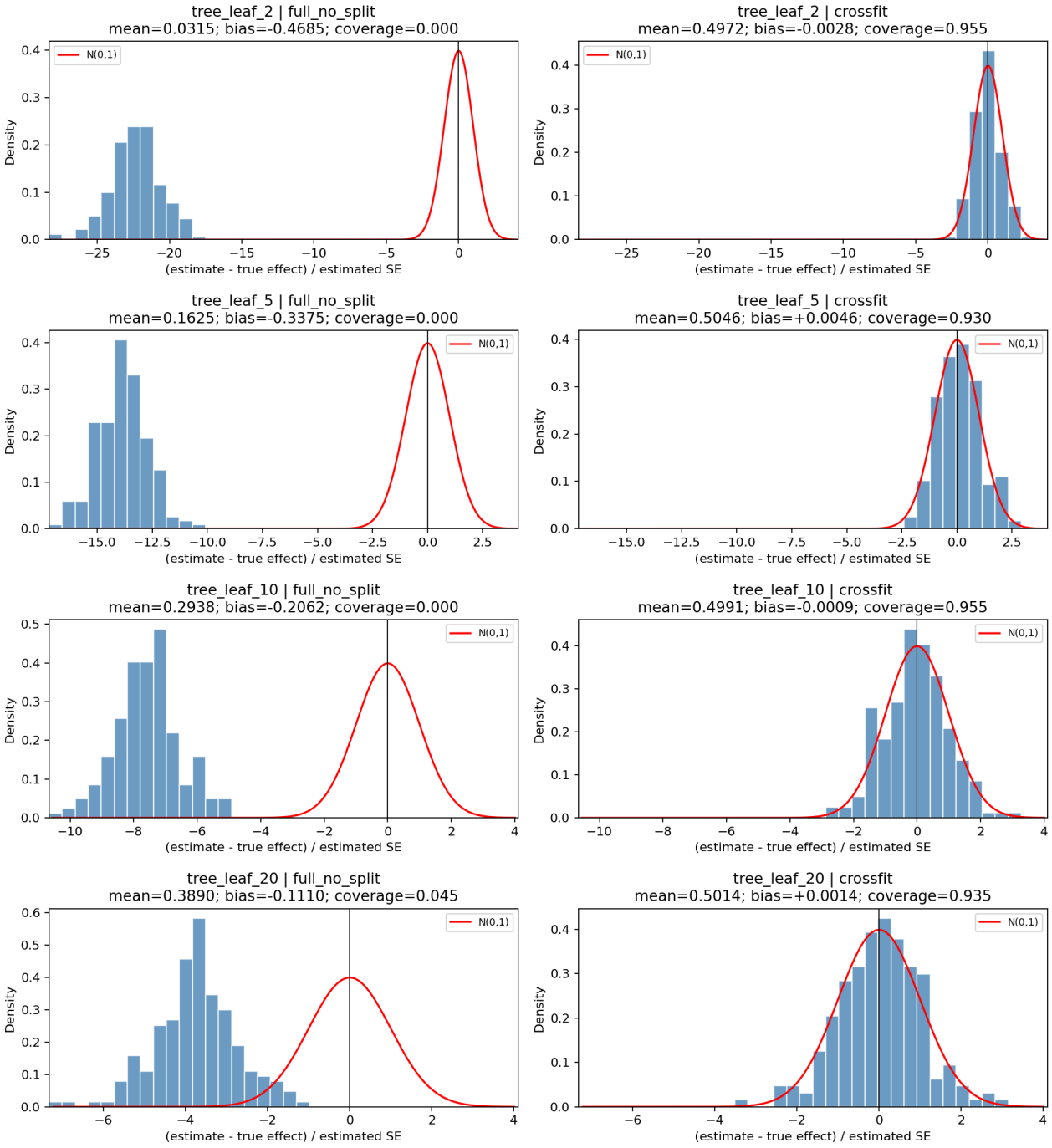
Table 9: Controlled tree cross-fit settings

| DGP | Tree min. leaf | Mean estimate | Bias | Coverage |
|---|---|---|---|---|
| Linear | 2 | 0.4984 | -0.0016 | 95.5% |
| Linear | 5 | 0.4996 | -0.0004 | 93.5% |
| Linear | 10 | 0.5013 | +0.0013 | 95.0% |
| Linear | 20 | 0.5007 | +0.0007 | 92.5% |
| Nonlinear | 2 | 0.4972 | -0.0028 | 95.5% |
| Nonlinear | 5 | 0.5046 | +0.0046 | 93.0% |
| Nonlinear | 10 | 0.4991 | -0.0009 | 95.5% |
| Nonlinear | 20 | 0.5014 | +0.0014 | 93.5% |

Across these eight tree-by-DGP settings, maximum absolute MC mean bias is below 0.0046. This descriptive bound is not a guarantee that every mean-bias interval lies inside +/-0.01. The broader controlled and ordinary-learner outcomes remain in the full 106-cell nuisance summary.




Controlled nuisance study (linear DGP). Original saved figure, byte-for-byte unchanged.




Controlled nuisance study (nonlinear DGP). Original saved figure, byte-for-byte unchanged.

## 7. Sensitivity analysis and limitations

The fixed grid varies n=300/1,000/3,000, K=2/5, baseline/moderate DGP level and four one-nuisance-role-at-a-time regularization alternatives. All 600 reported summary cells are retained, with paired MAE/MSE tests, Holm corrections and error decompositions. This is not a full factorial hyperparameter grid; some cells are aliases, not independent runs.

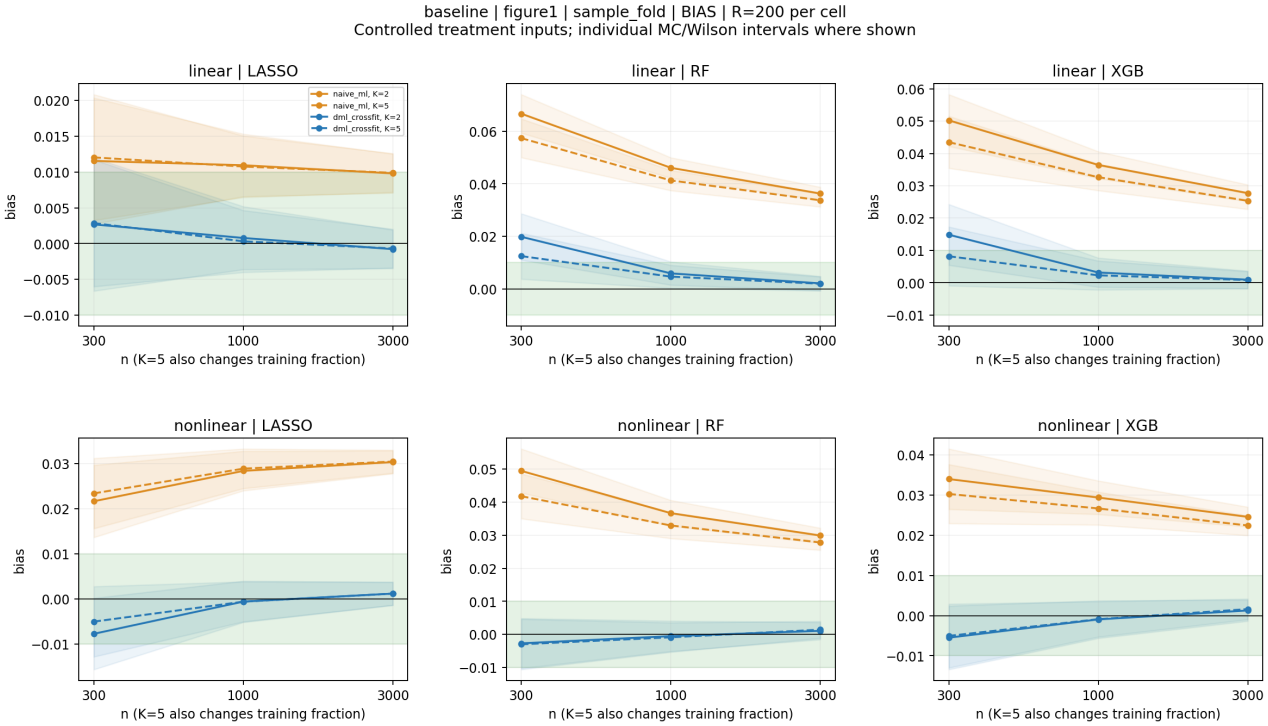
Table 10: Cross-fit ranges over the complete sensitivity grid

| Design | Learner | Bias range | Coverage range |
|---|---|---|---|
| Orthogonalization | Random Forest | [-0.0044, +0.0305] | [88.0%, 96.5%] |
| Orthogonalization | XGBoost | [-0.0098, +0.0150] | [92.0%, 98.0%] |
| Orthogonalization | Lasso | [-0.0096, +0.0029] | [91.5%, 97.0%] |
| Cross-fitting | Random Forest | [-0.0150, +0.0217] | [87.5%, 97.5%] |
| Cross-fitting | XGBoost | [-0.0169, +0.0048] | [90.0%, 97.5%] |
| Cross-fitting | Lasso | [-0.0012, +0.0073] | [90.5%, 96.5%] |

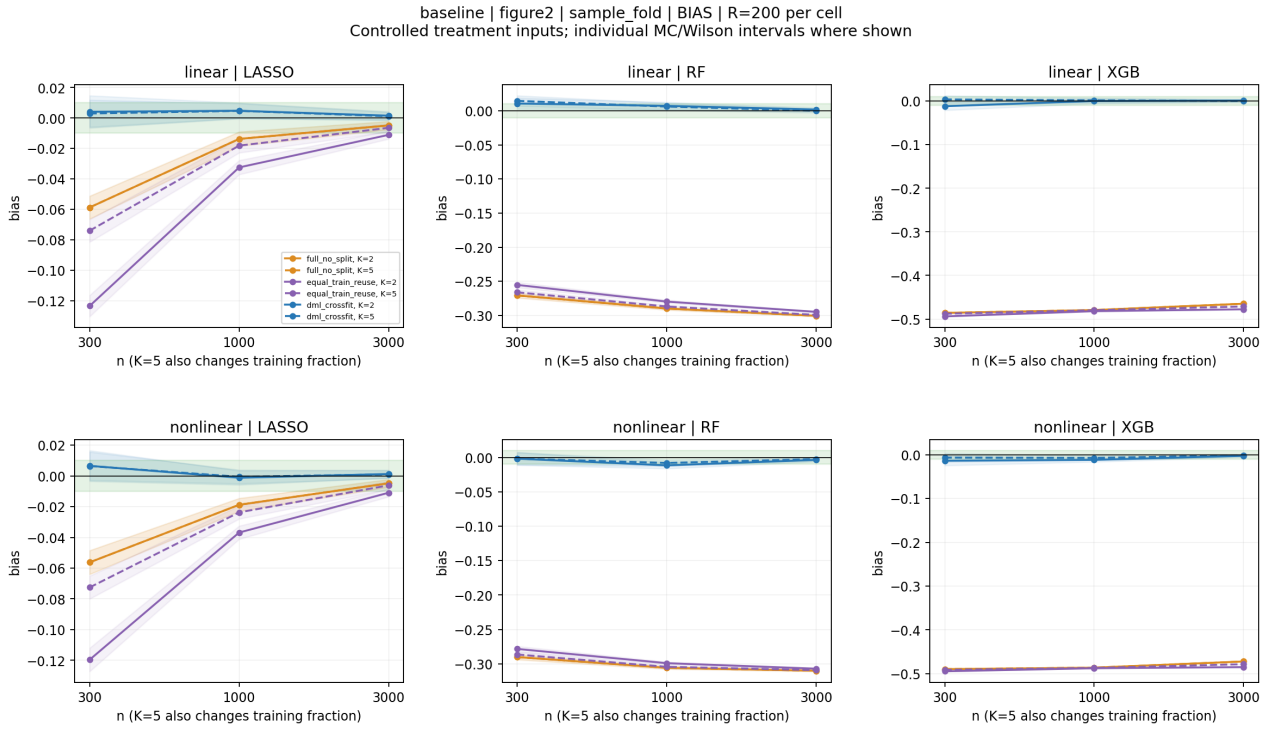
Ranges describe heterogeneous settings, not one pooled estimator. Coverage can fall to 87.5%; five folds are not uniformly superior to two; and lower absolute mean bias need not lower MAE or RMSE. The sensitivity study evaluates robustness boundaries instead of proving unconditional robustness.

# Baseline sensitivity: sample fold / BIAS

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



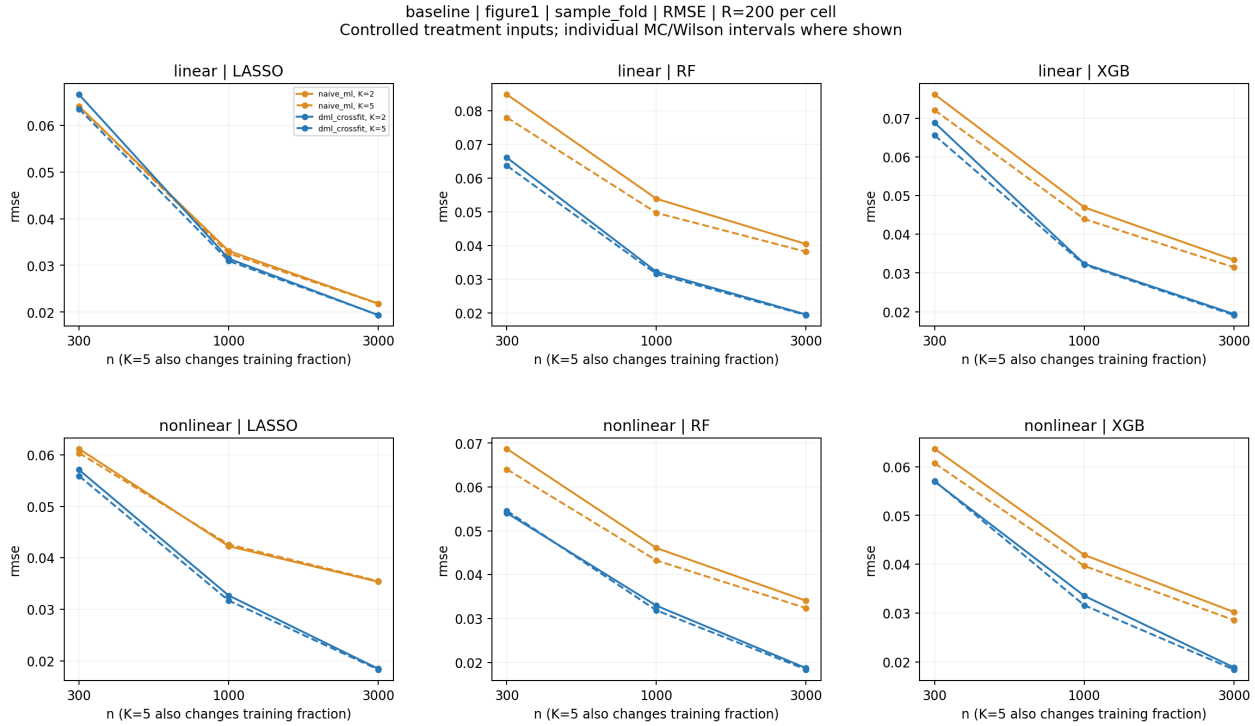

baseline figure1 sample fold bias. Original saved figure, unchanged.



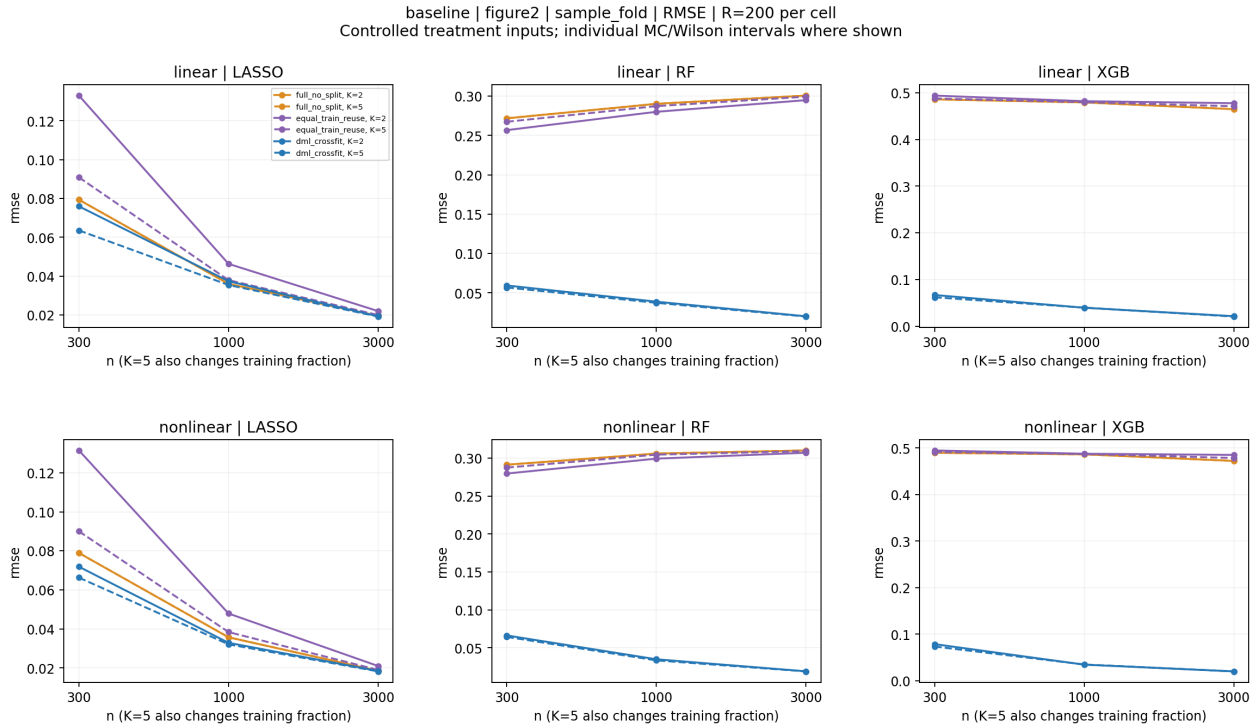

baseline figure2 sample fold bias. Original saved figure, unchanged.

# Baseline sensitivity: sample fold / RMSE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



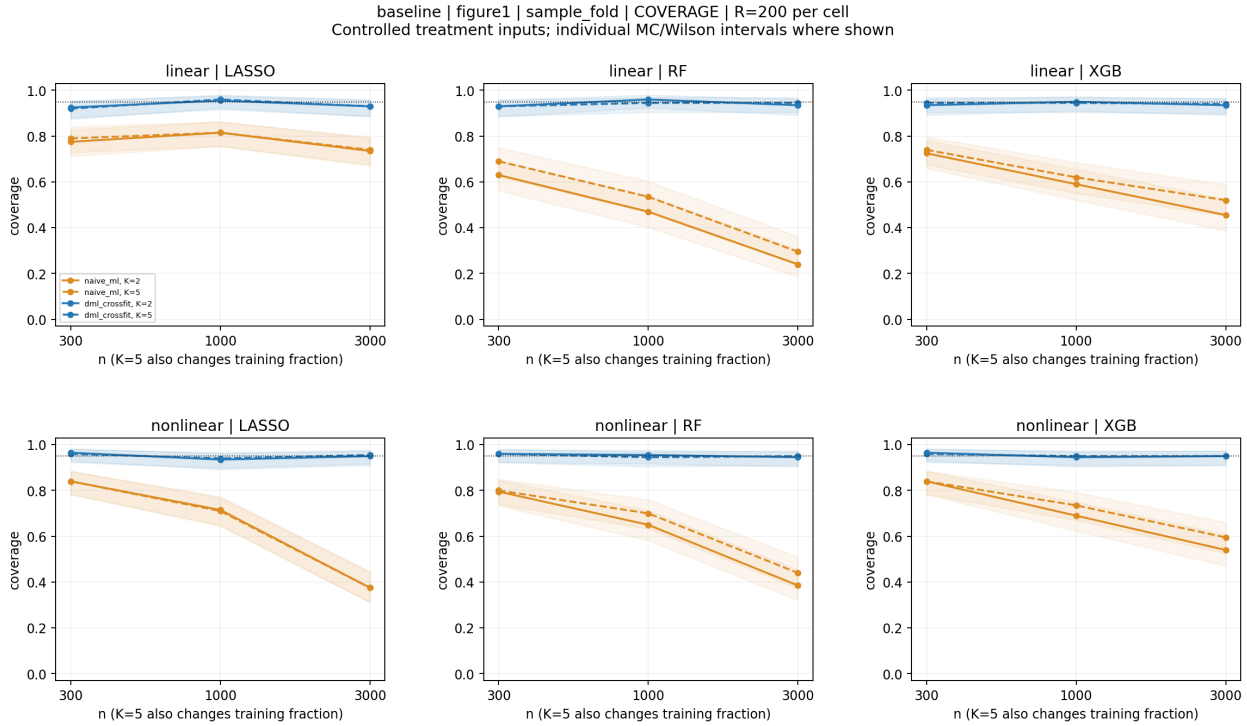

baseline figure1 sample fold rmse. Original saved figure, unchanged.



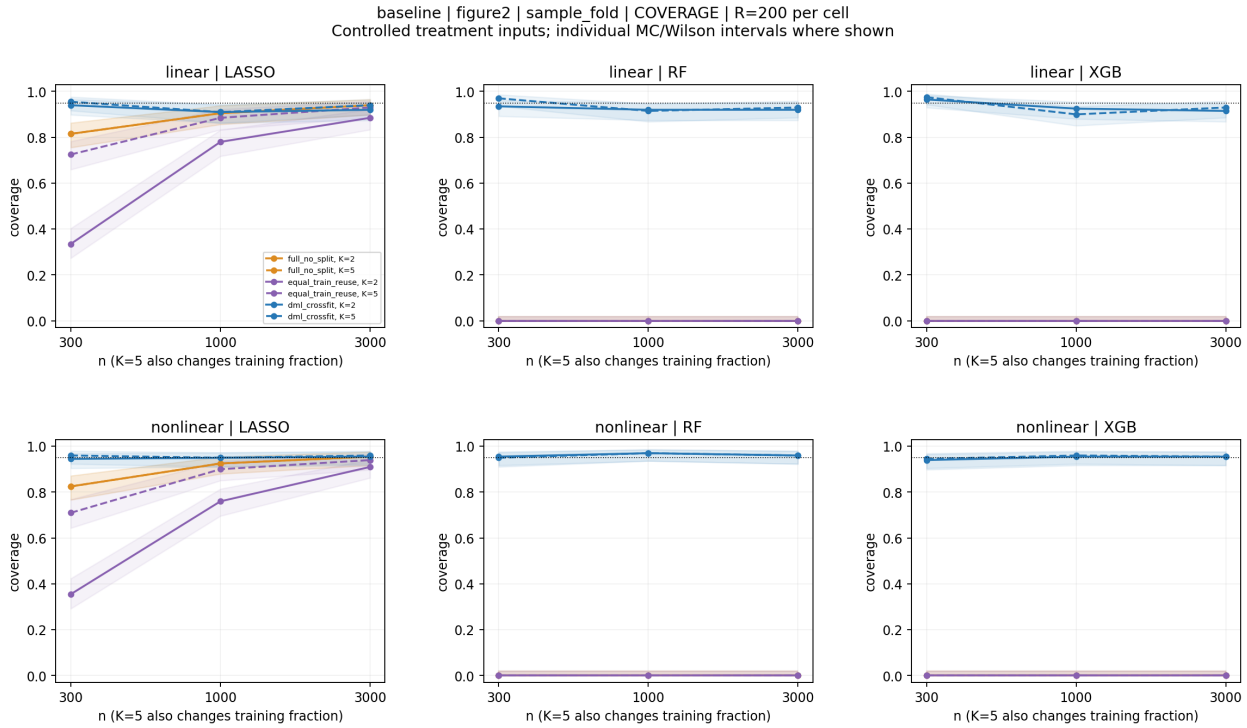

baseline figure2 sample fold rmse. Original saved figure, unchanged.

# Baseline sensitivity: sample fold / COVERAGE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



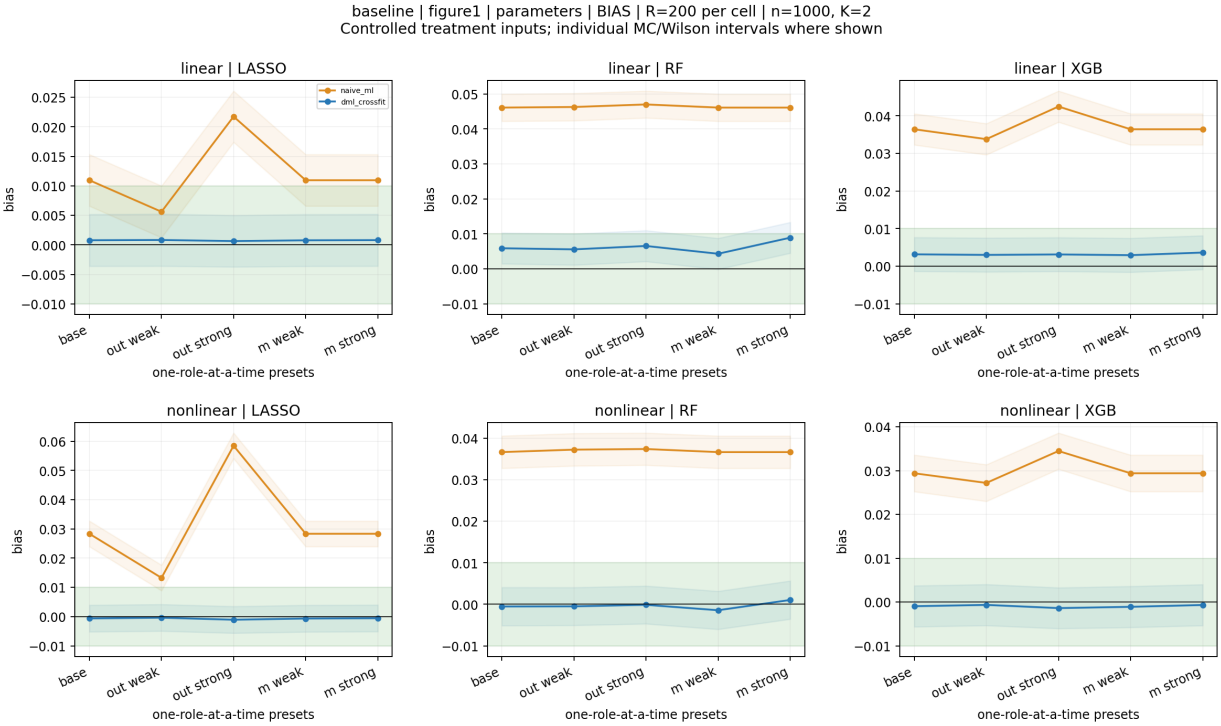

baseline figure1 sample fold coverage. Original saved figure, unchanged.



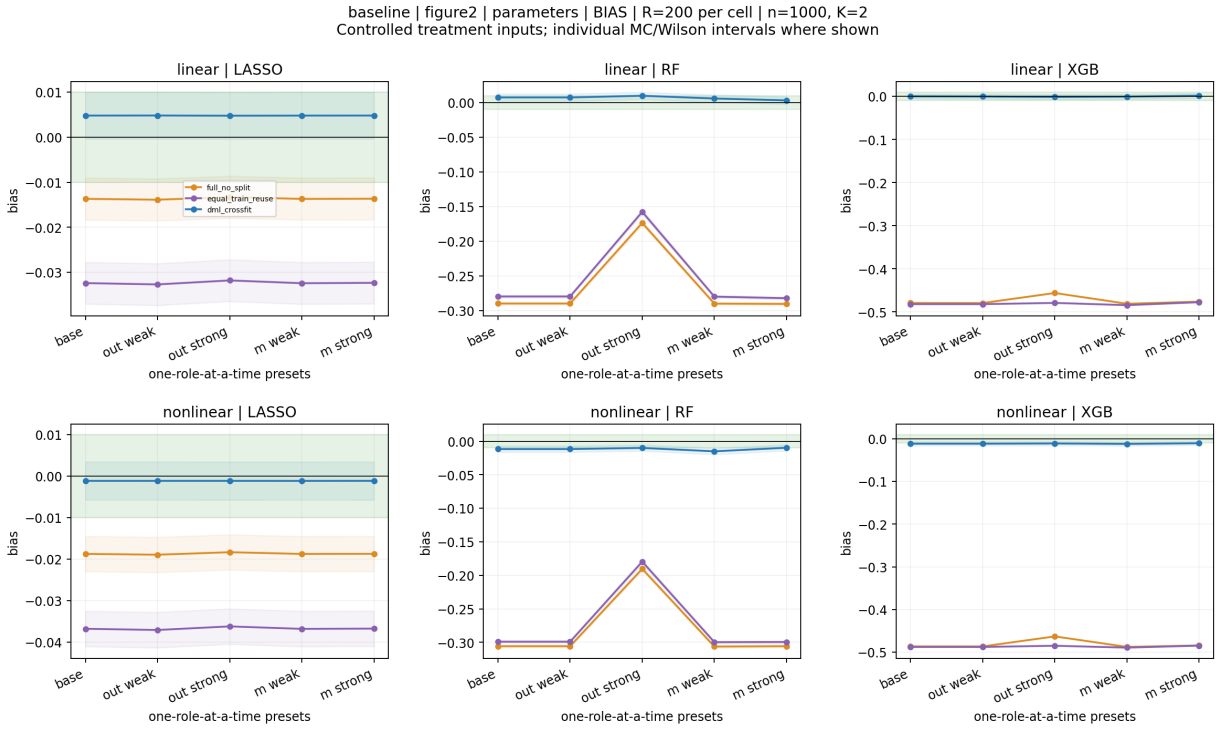

baseline figure2 sample fold coverage. Original saved figure, unchanged.

# Baseline sensitivity: parameters / BIAS

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



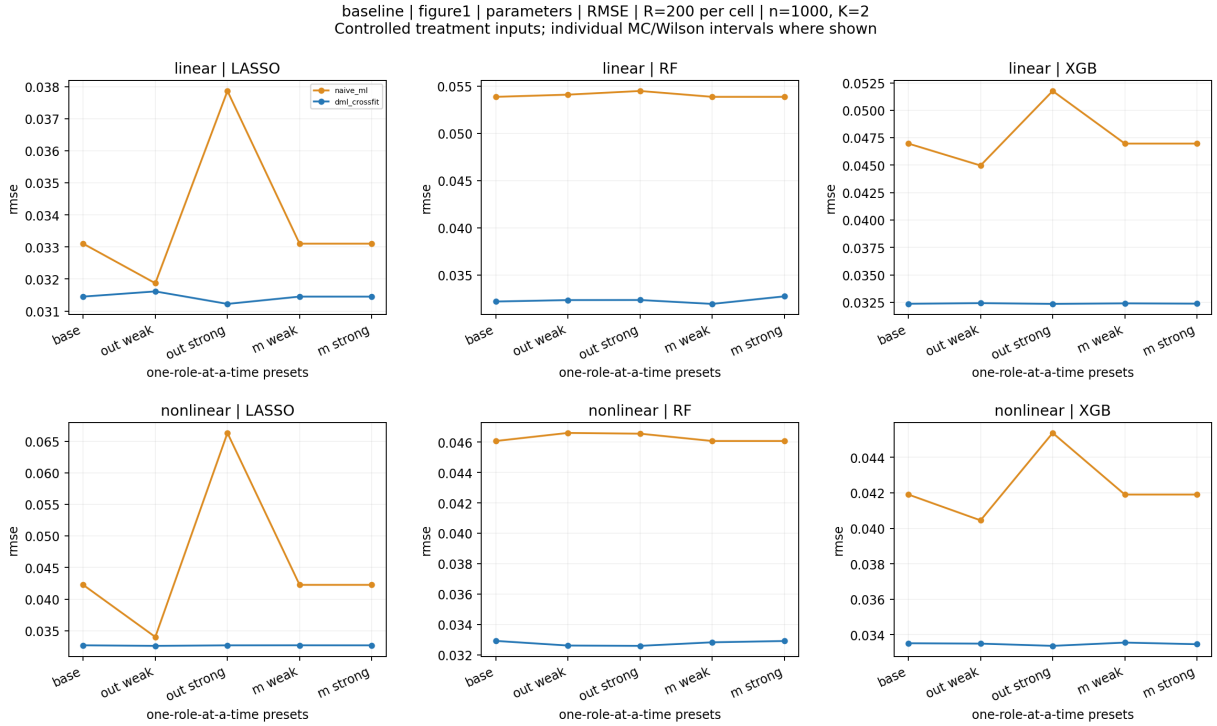

baseline figure1 parameters bias. Original saved figure, unchanged.



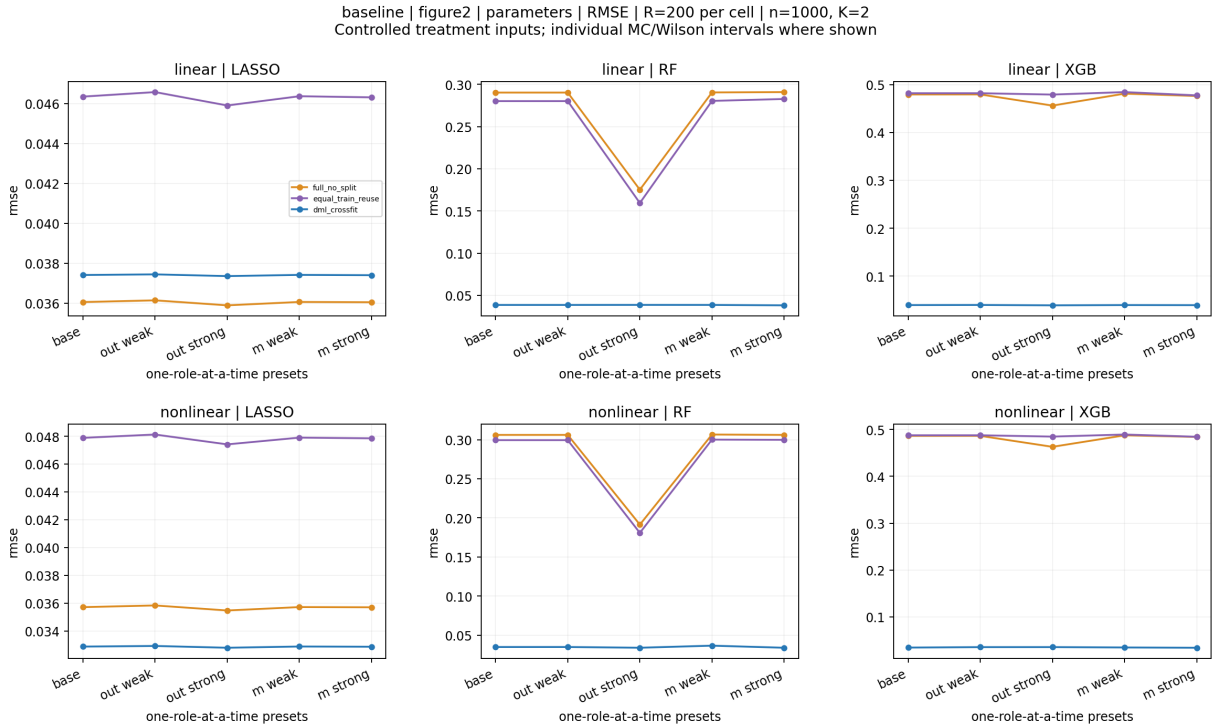

baseline figure2 parameters bias. Original saved figure, unchanged.

# Baseline sensitivity: parameters / RMSE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



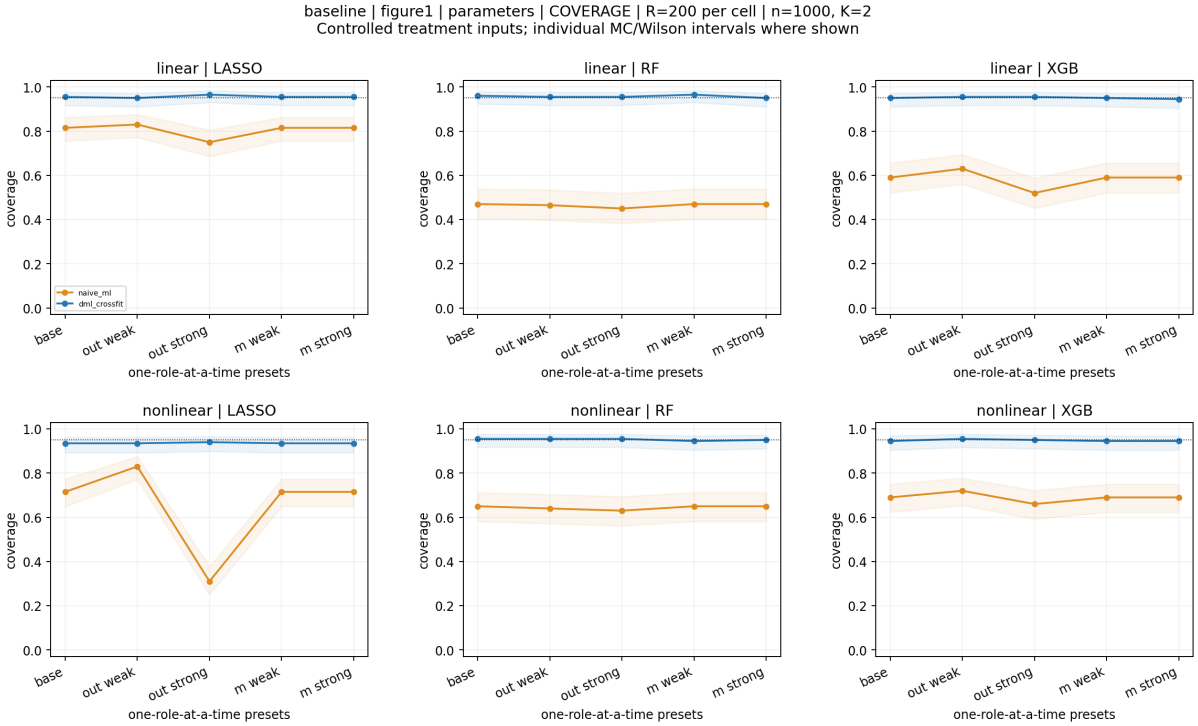

baseline figure1 parameters rmse. Original saved figure, unchanged.



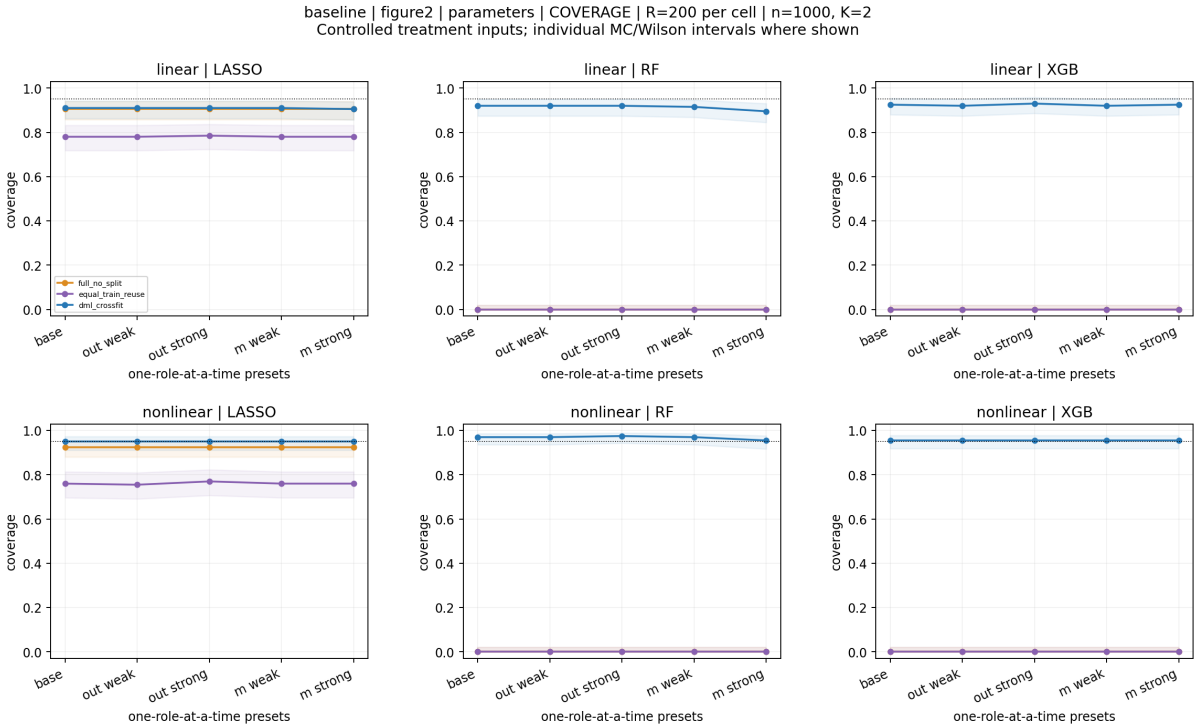

baseline figure2 parameters rmse. Original saved figure, unchanged.

# Baseline sensitivity: parameters / COVERAGE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



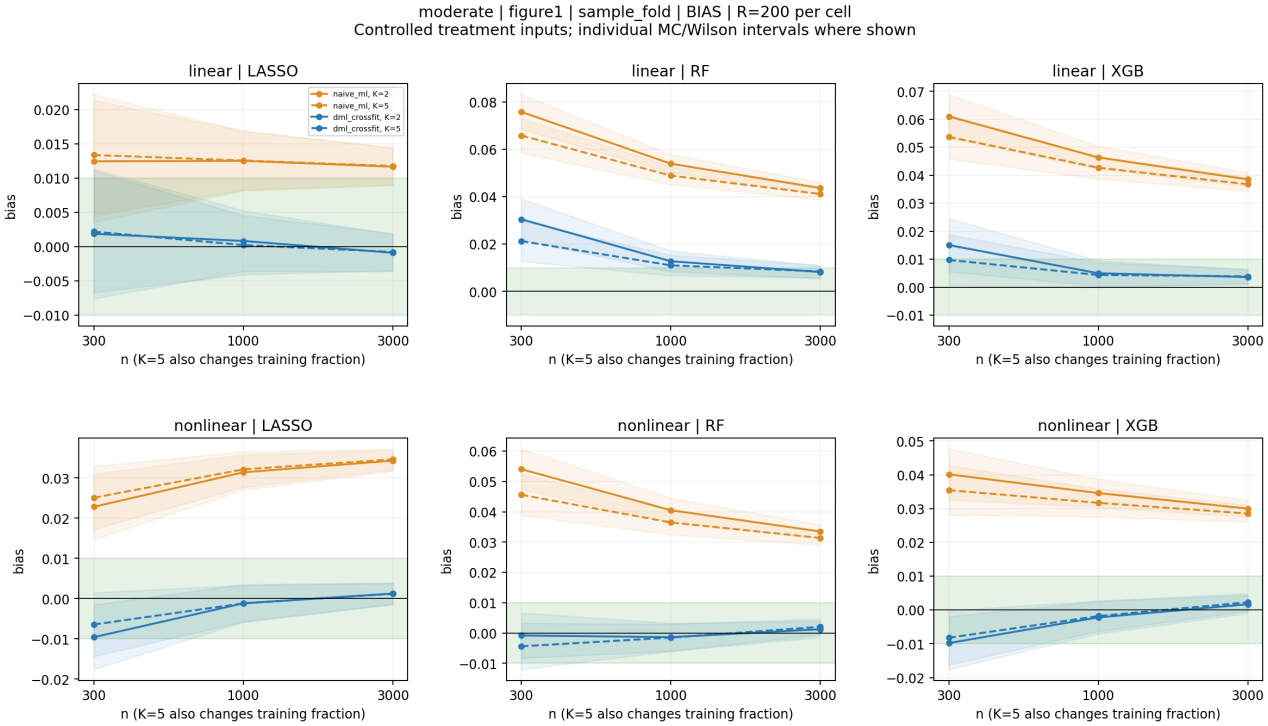

baseline figure1 parameters coverage. Original saved figure, unchanged.



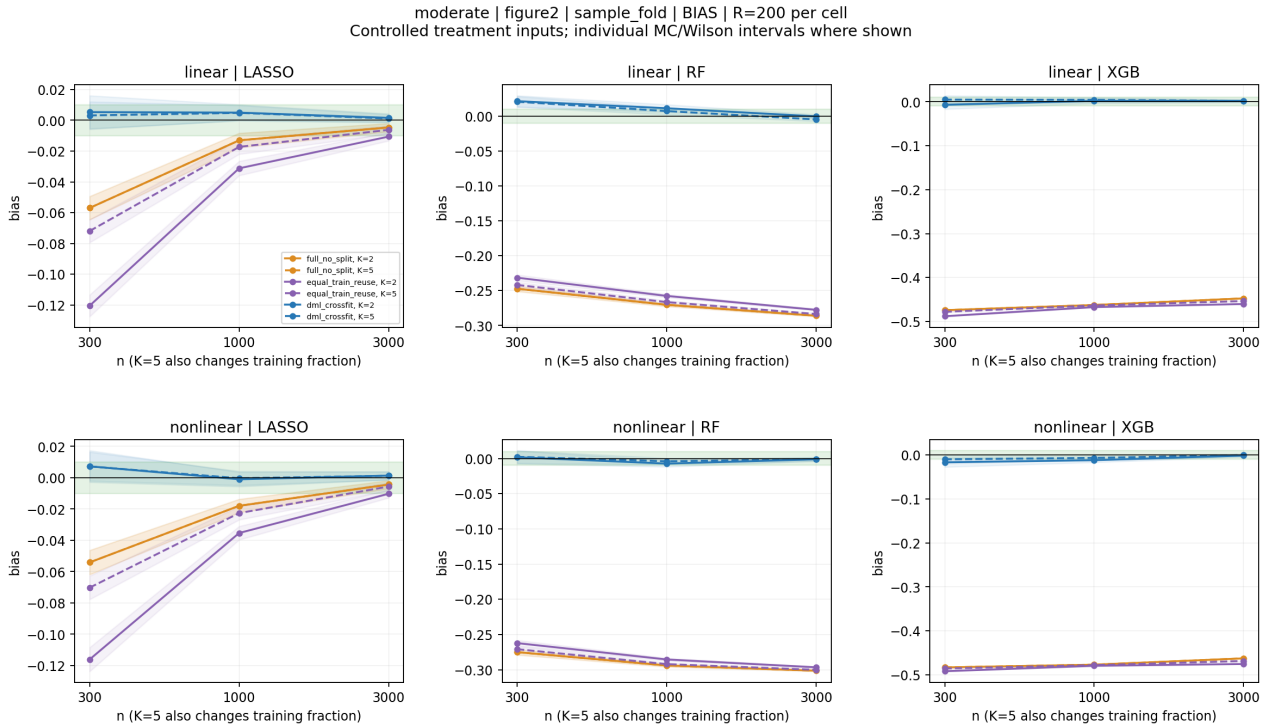

baseline figure2 parameters coverage. Original saved figure, unchanged.

# Moderate sensitivity: sample fold / BIAS

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



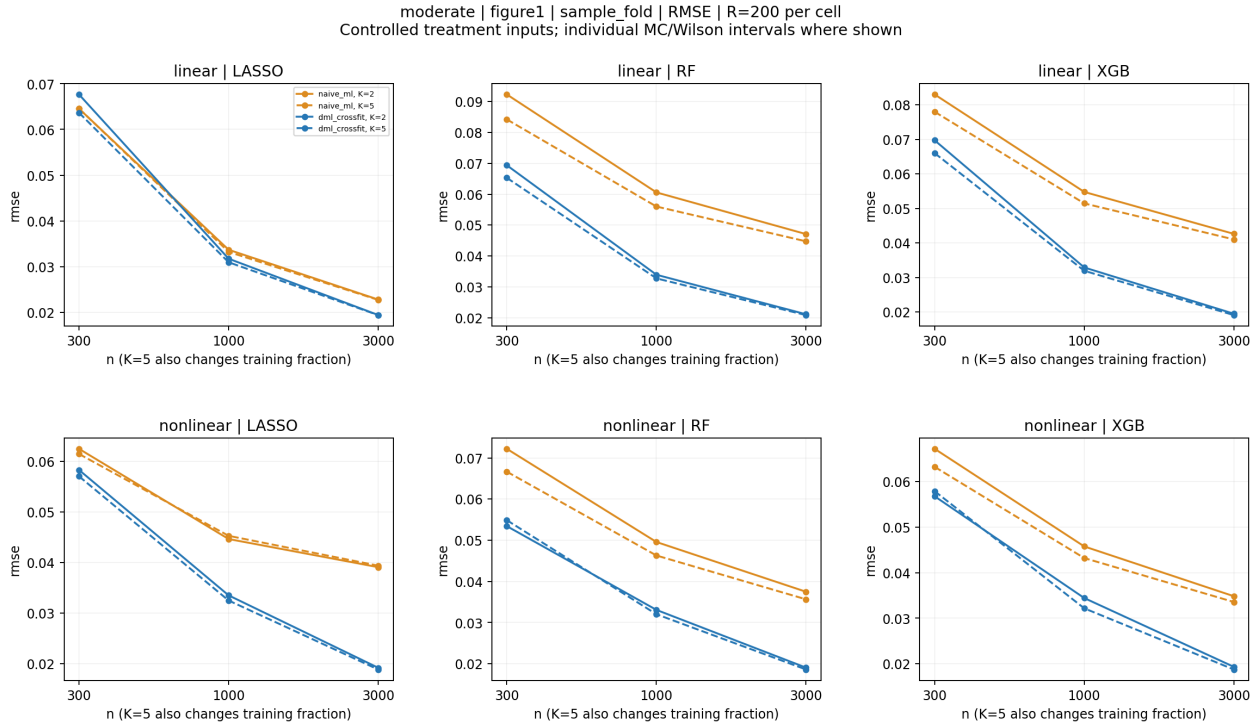

moderate figure1 sample fold bias. Original saved figure, unchanged.



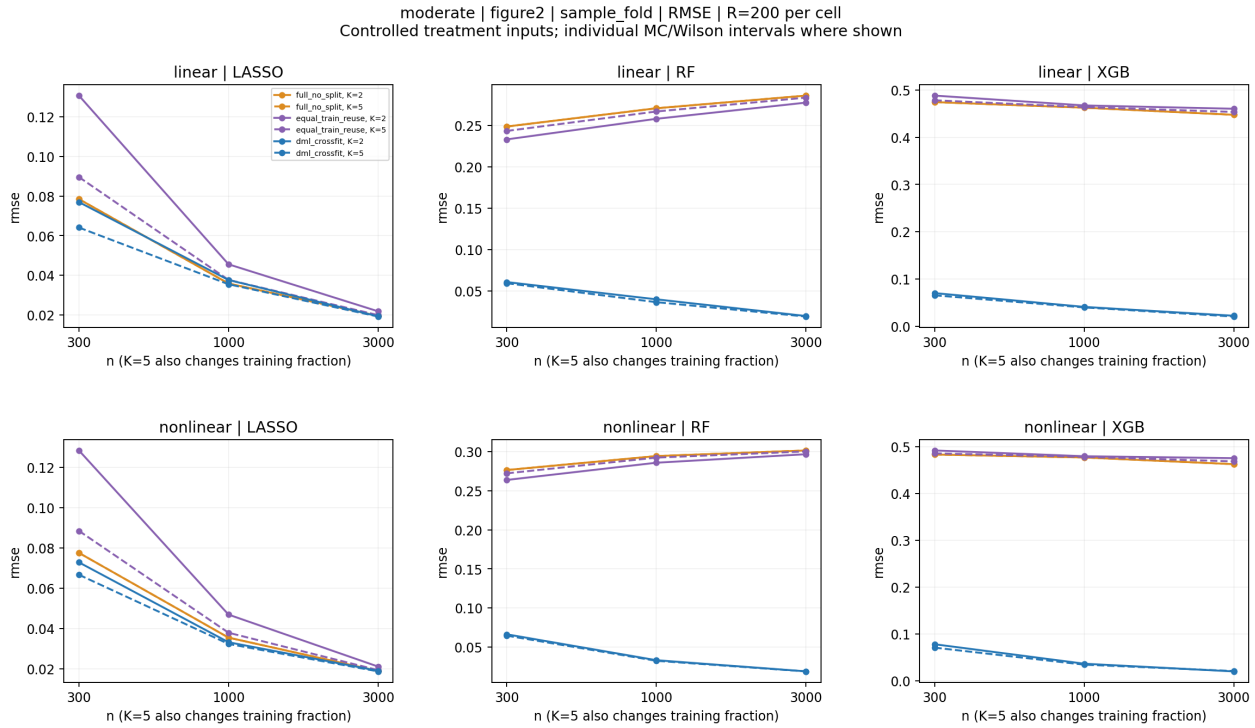

moderate figure2 sample fold bias. Original saved figure, unchanged.

# Moderate sensitivity: sample fold / RMSE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



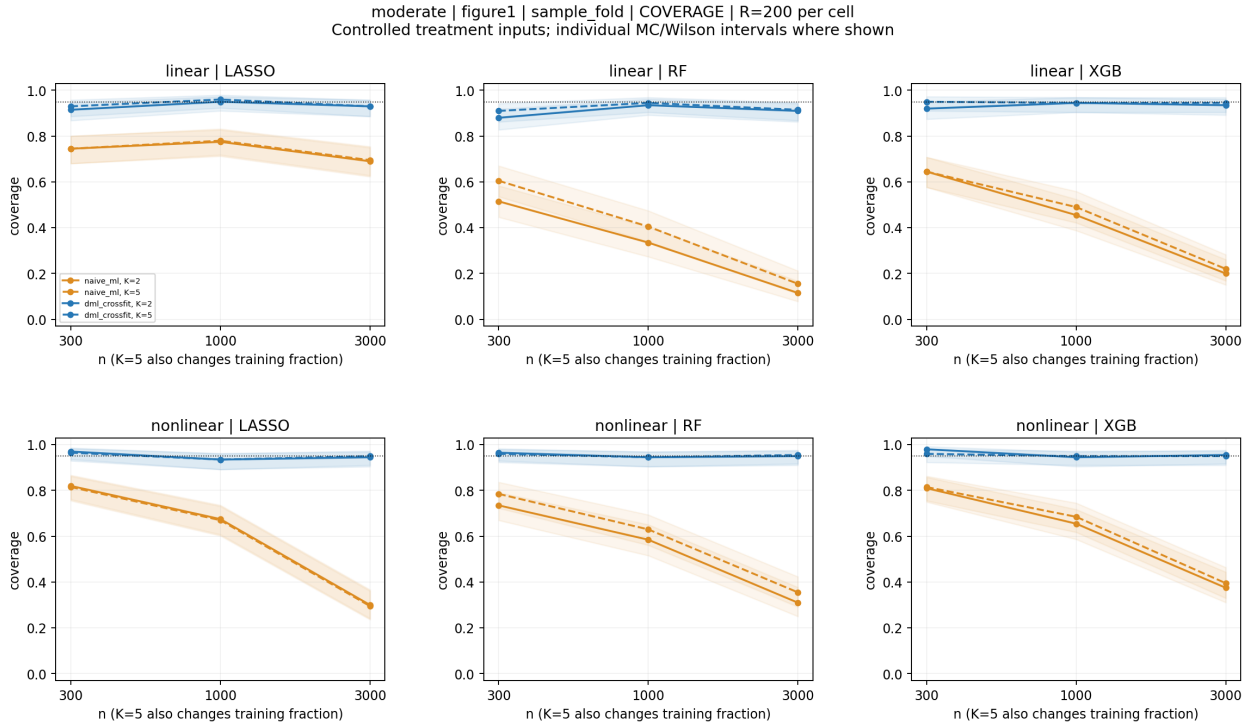

moderate figure1 sample fold rmse. Original saved figure, unchanged.



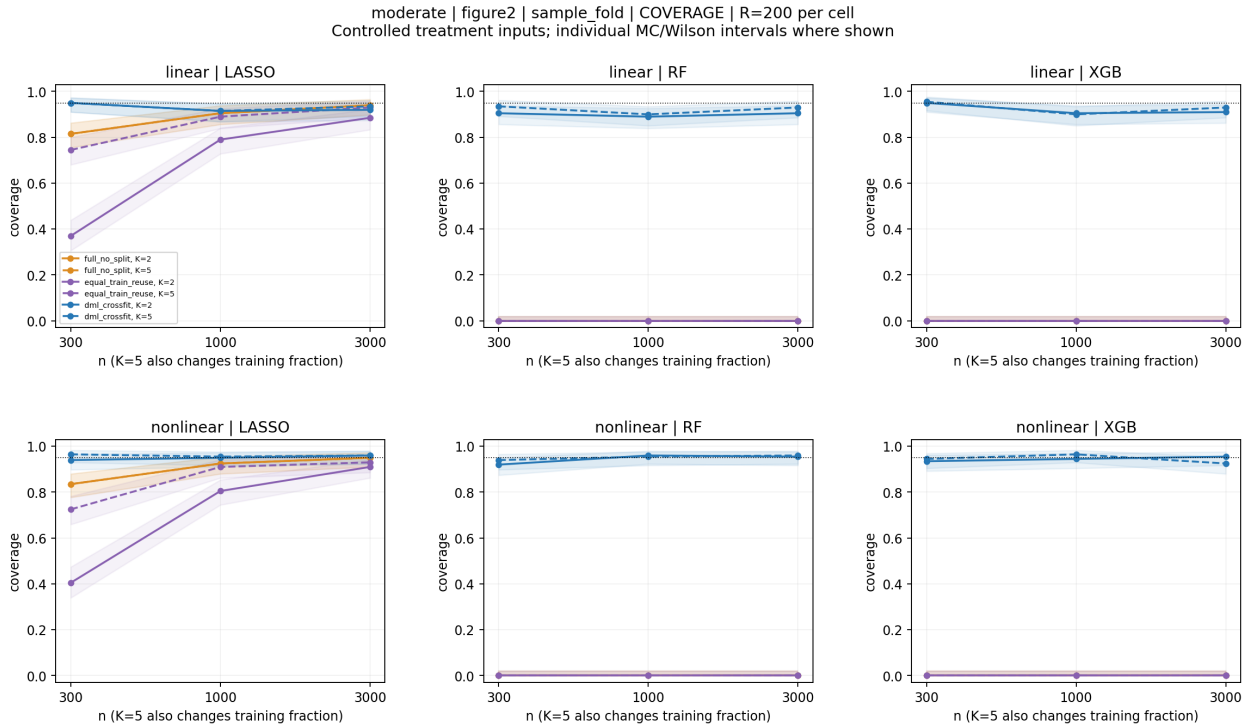

moderate figure2 sample fold rmse. Original saved figure, unchanged.

# Moderate sensitivity: sample fold / COVERAGE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



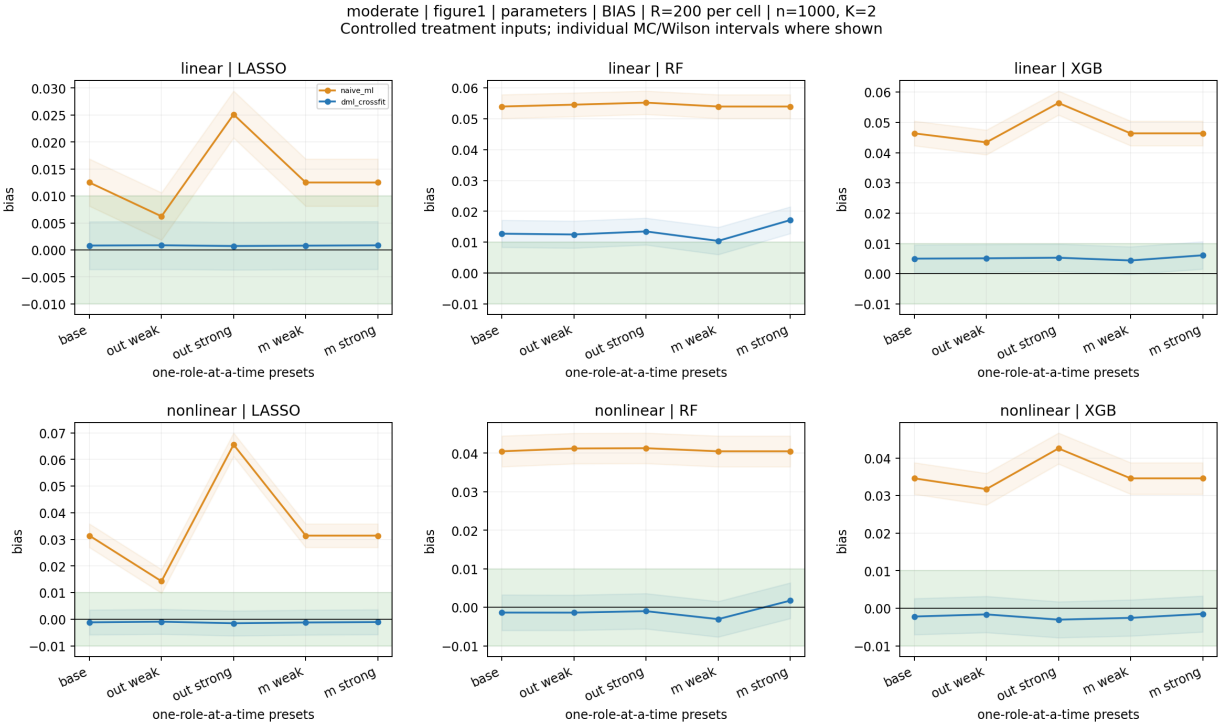

moderate figure1 sample fold coverage. Original saved figure, unchanged.



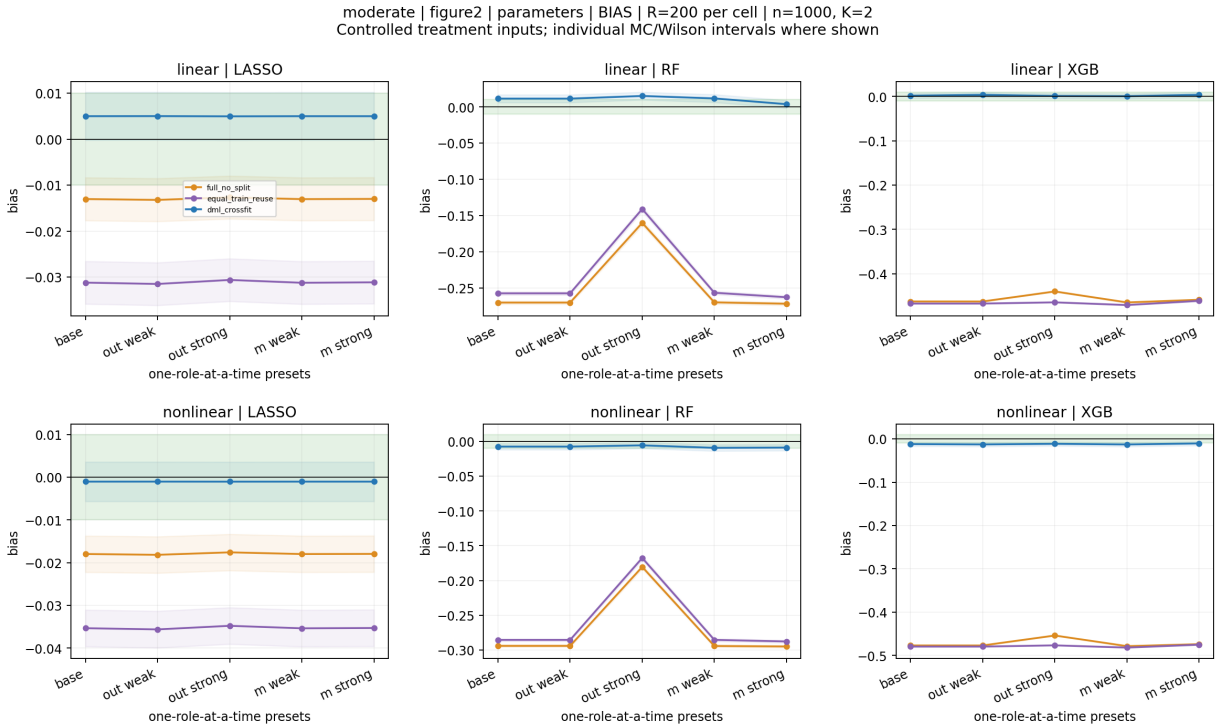

moderate figure2 sample fold coverage. Original saved figure, unchanged.

# Moderate sensitivity: parameters / BIAS

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



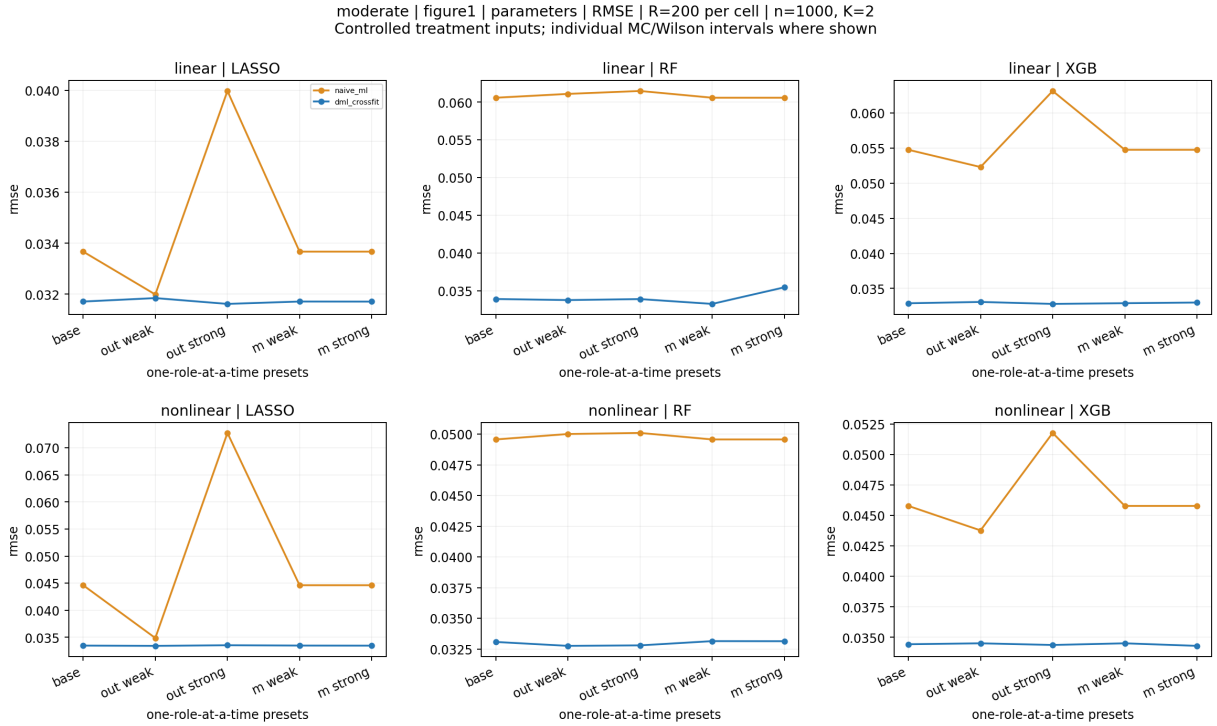

moderate figure1 parameters bias. Original saved figure, unchanged.



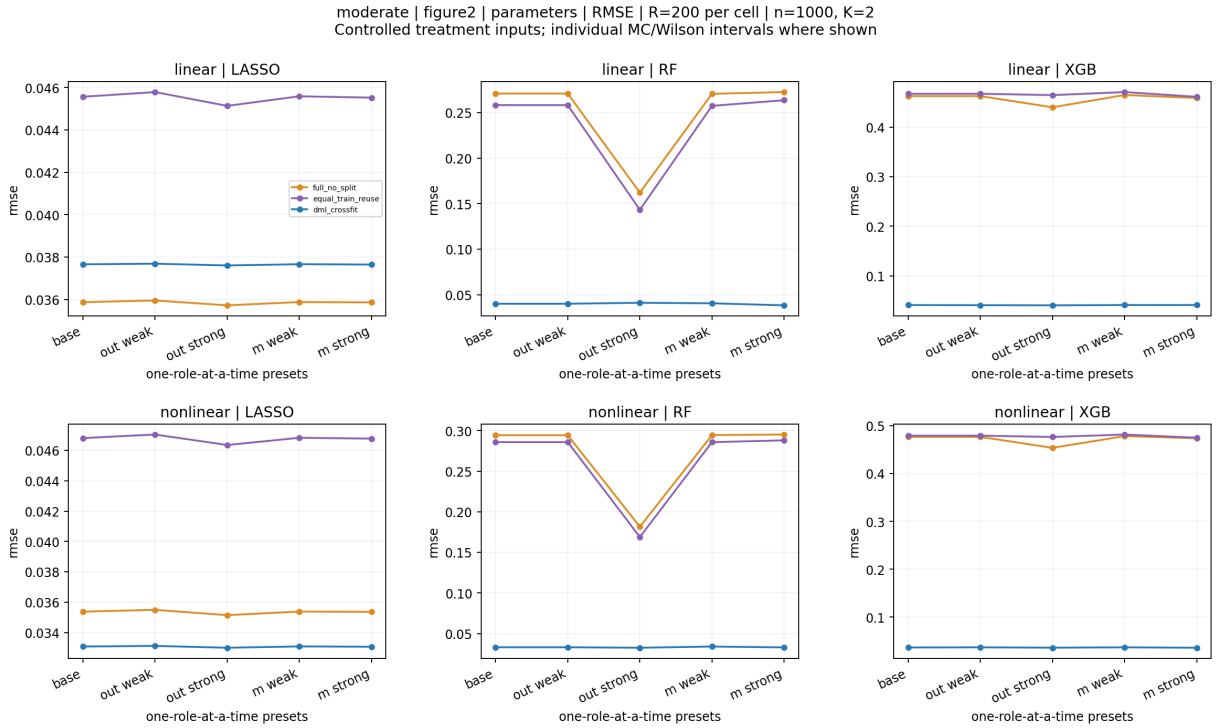

moderate figure2 parameters bias. Original saved figure, unchanged.

# Moderate sensitivity: parameters / RMSE

Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



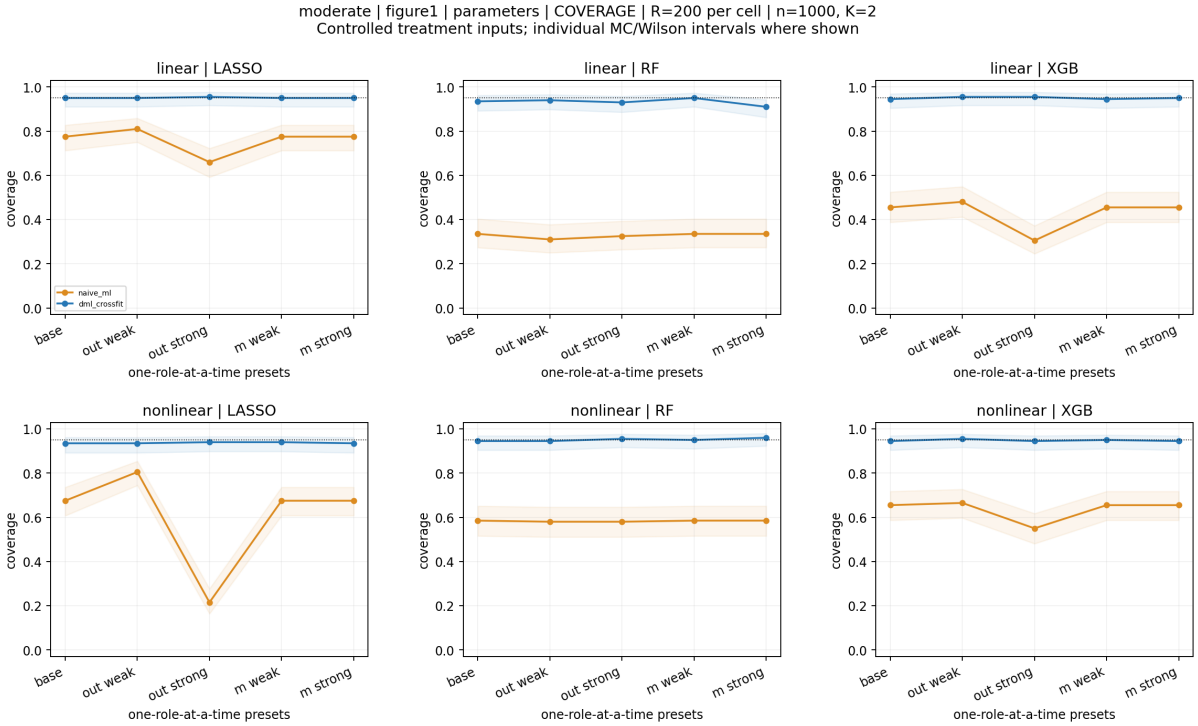

moderate figure1 parameters rmse. Original saved figure, unchanged.



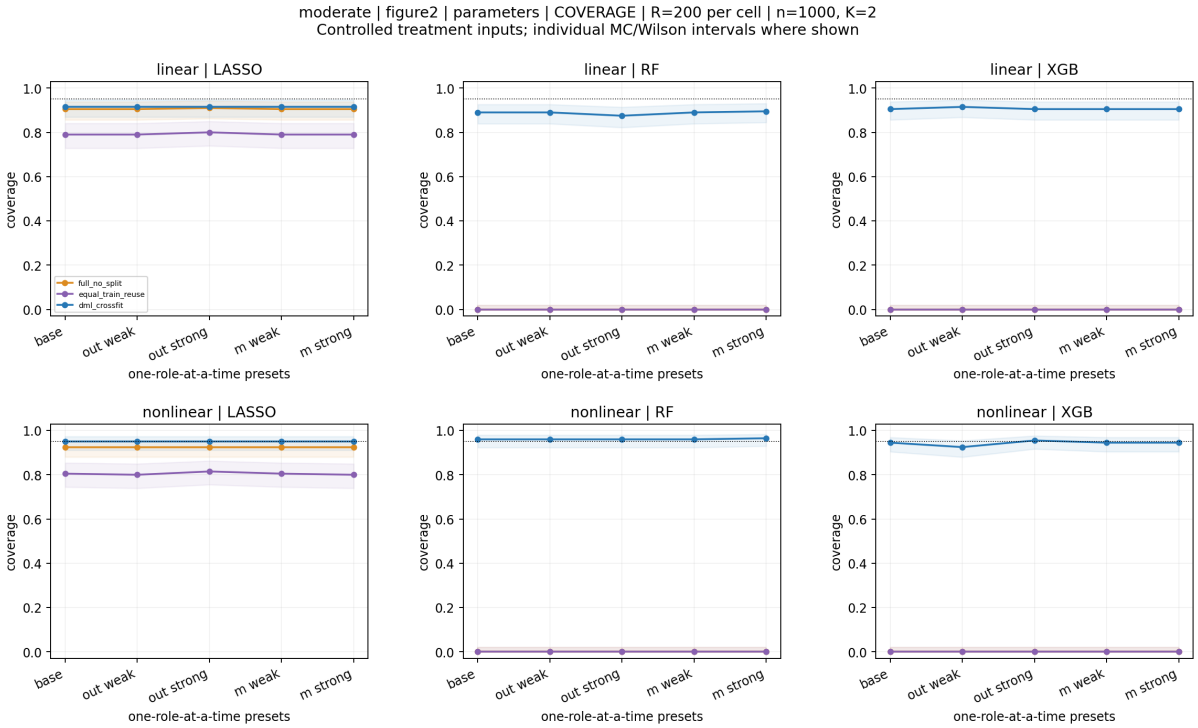

moderate figure2 parameters rmse. Original saved figure, unchanged.

# Moderate sensitivity: parameters / COVERAGE

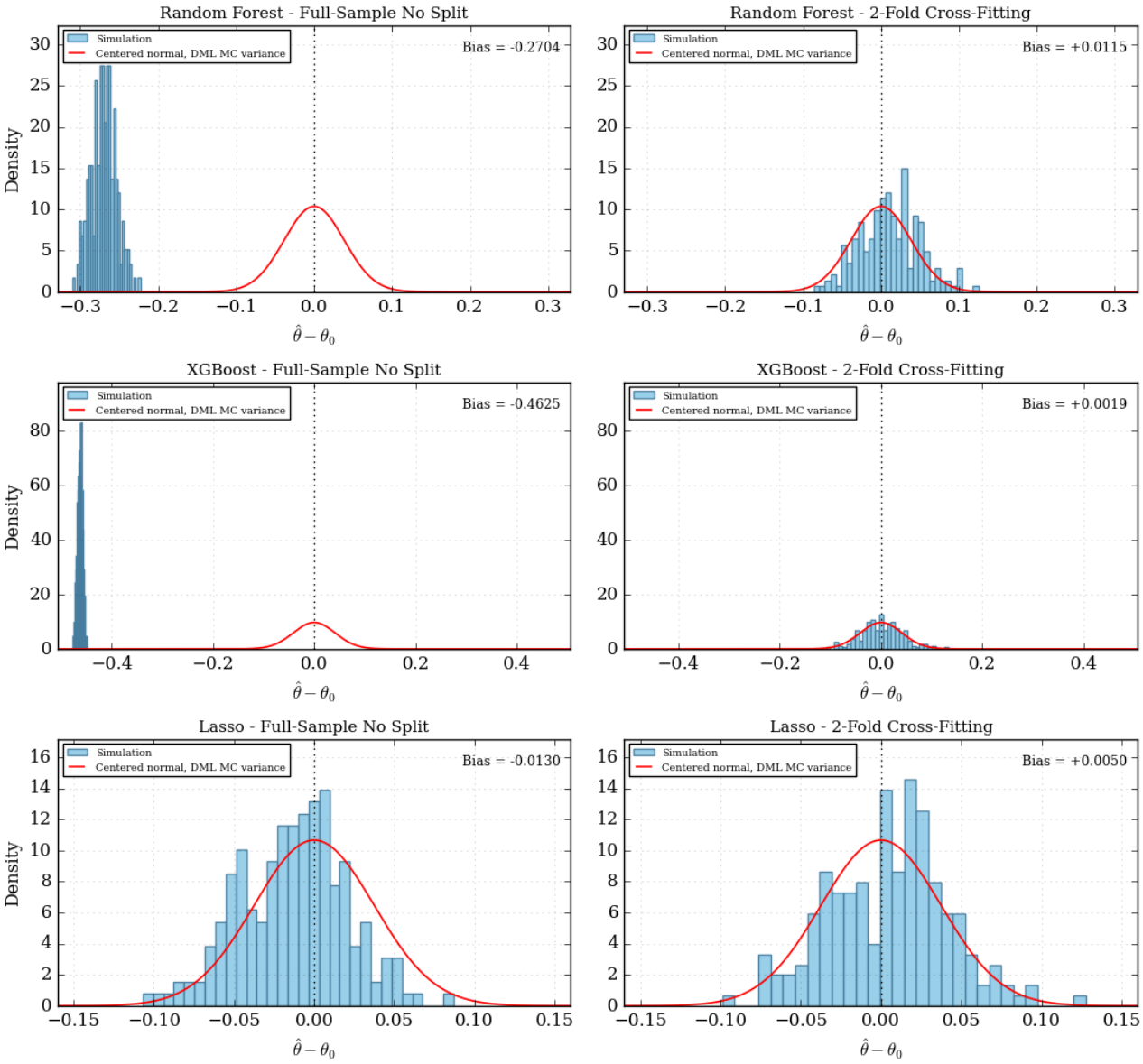
Each image contains BOTH DGPs: linear in the upper row and nonlinear in the lower row. These are separate validation settings, not additional observations pooled into the CV headline.



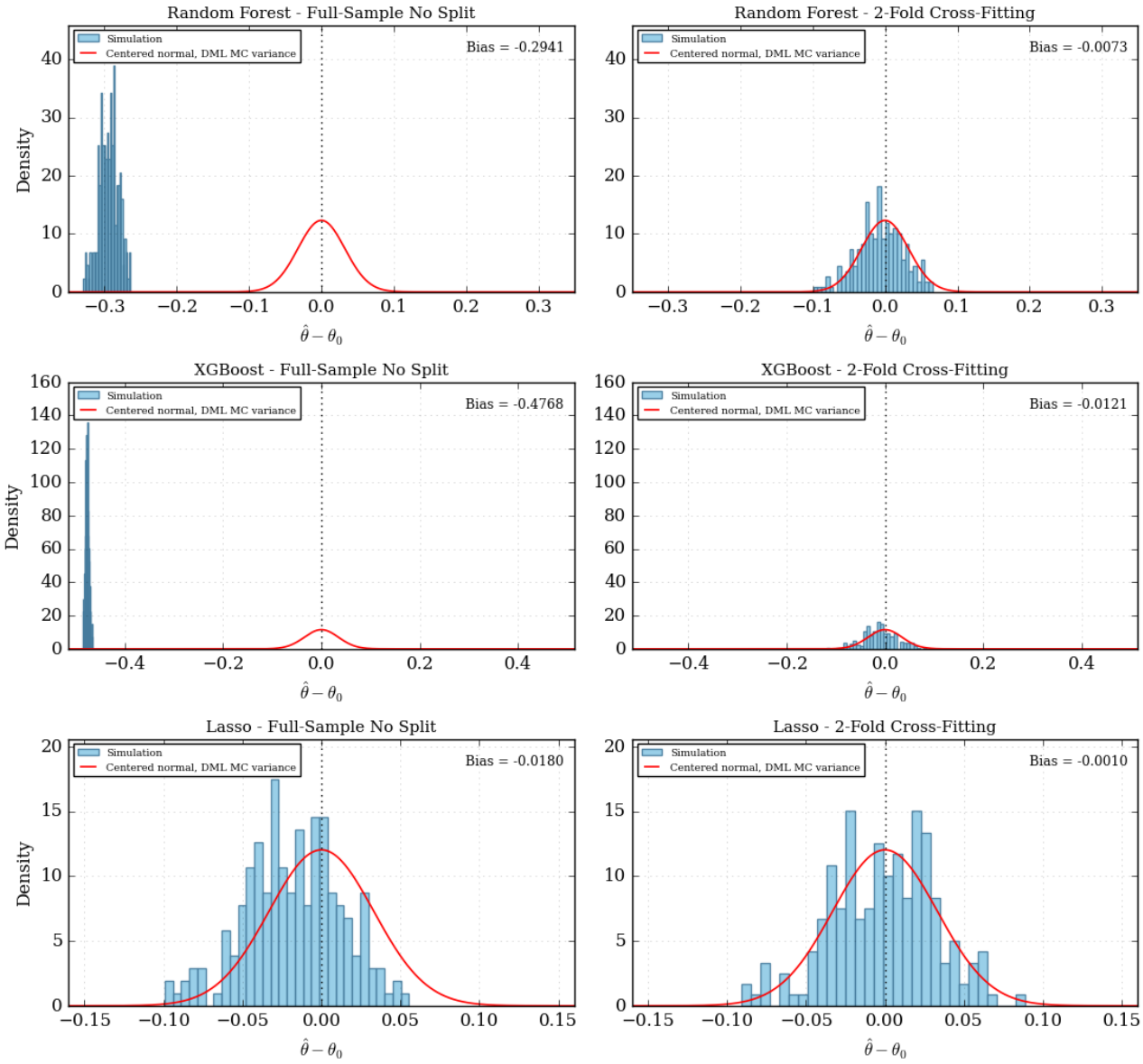

moderate figure1 parameters coverage. Original saved figure, unchanged.



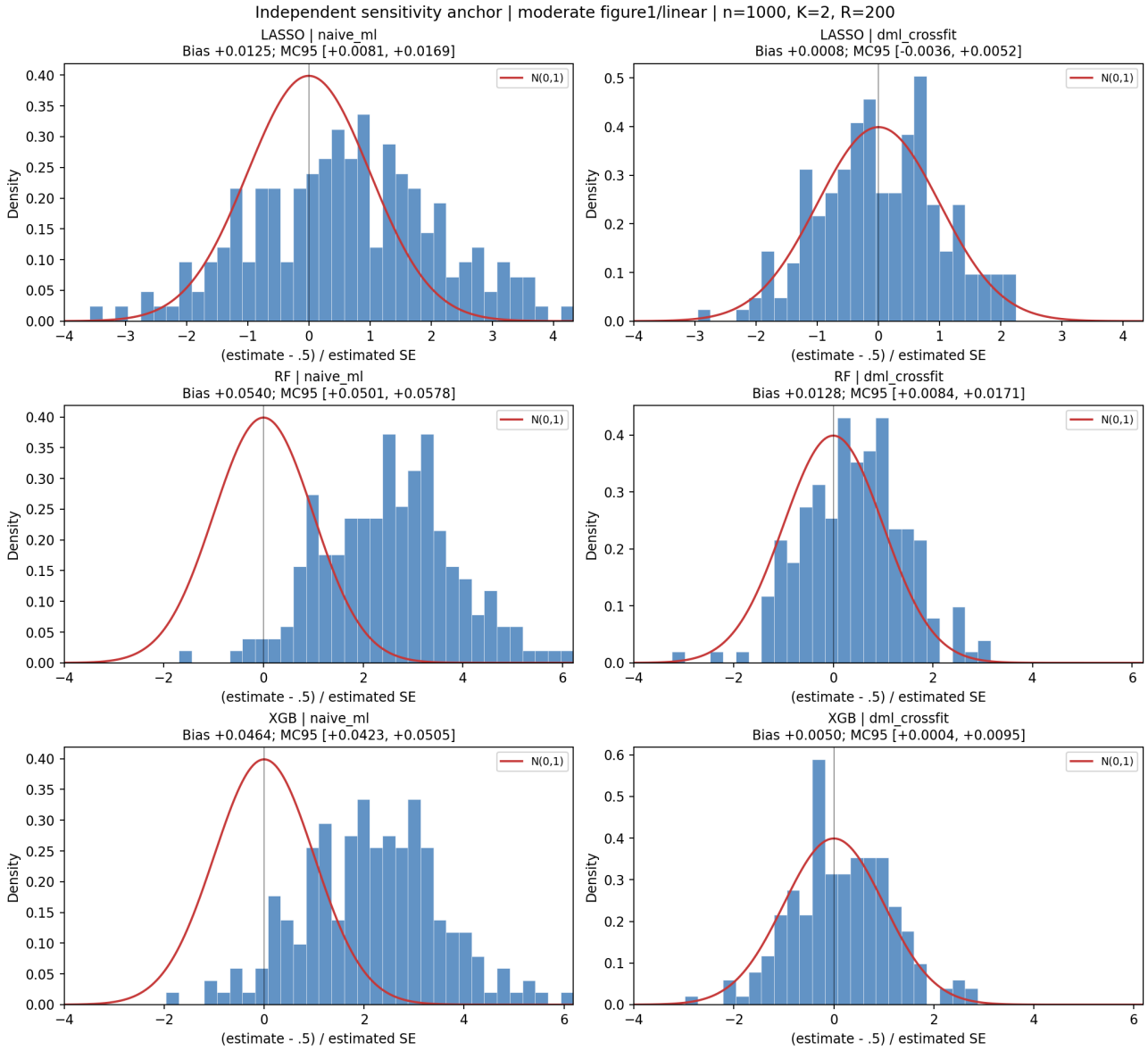

moderate figure2 parameters coverage. Original saved figure, unchanged.



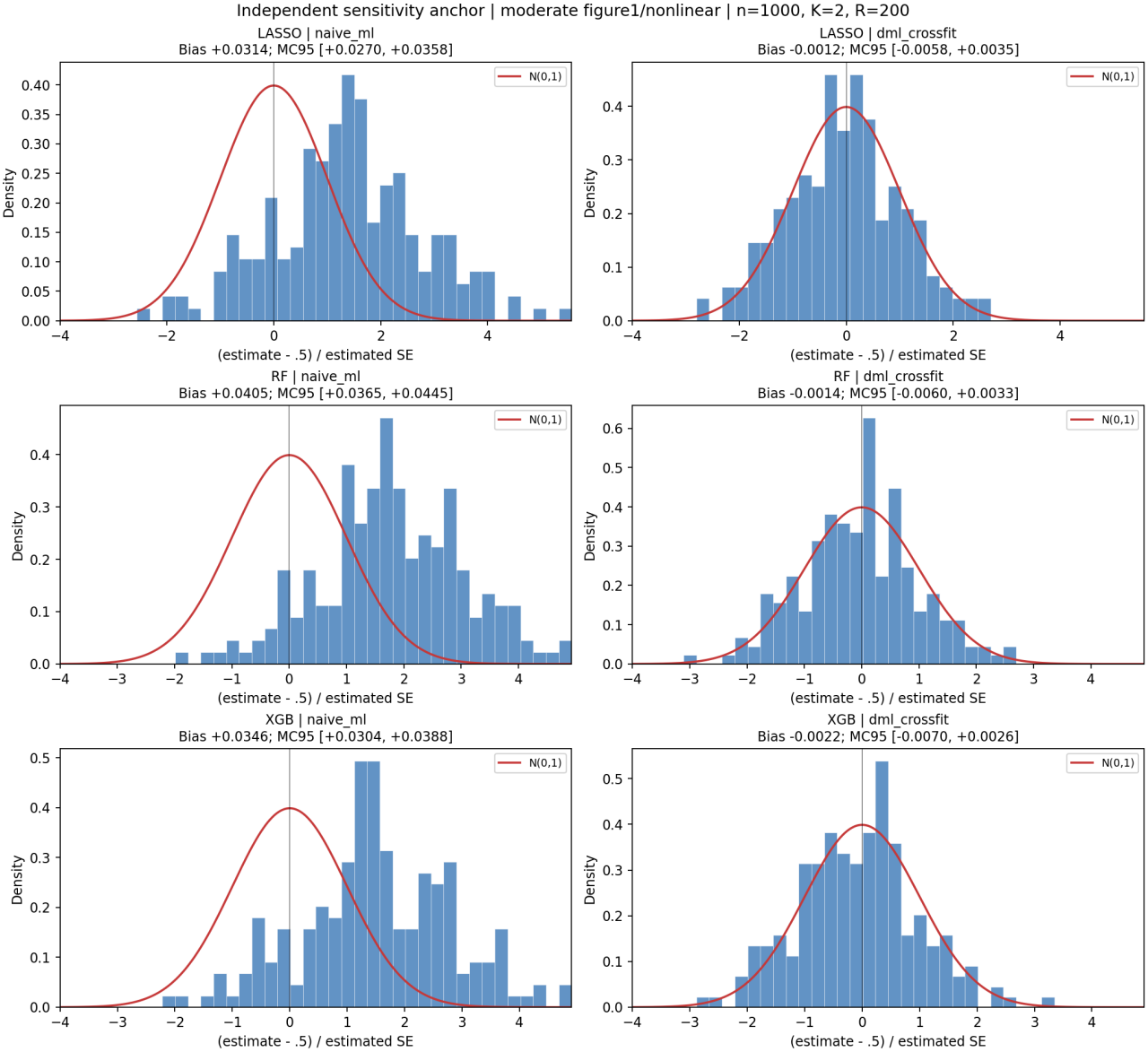

Raw-error companion: full no-split versus cross-fitting (linear moderate DGP, n=1,000, R=200). Every draw is shown on common within-learner axes. This plot is not a studentized-error plot.



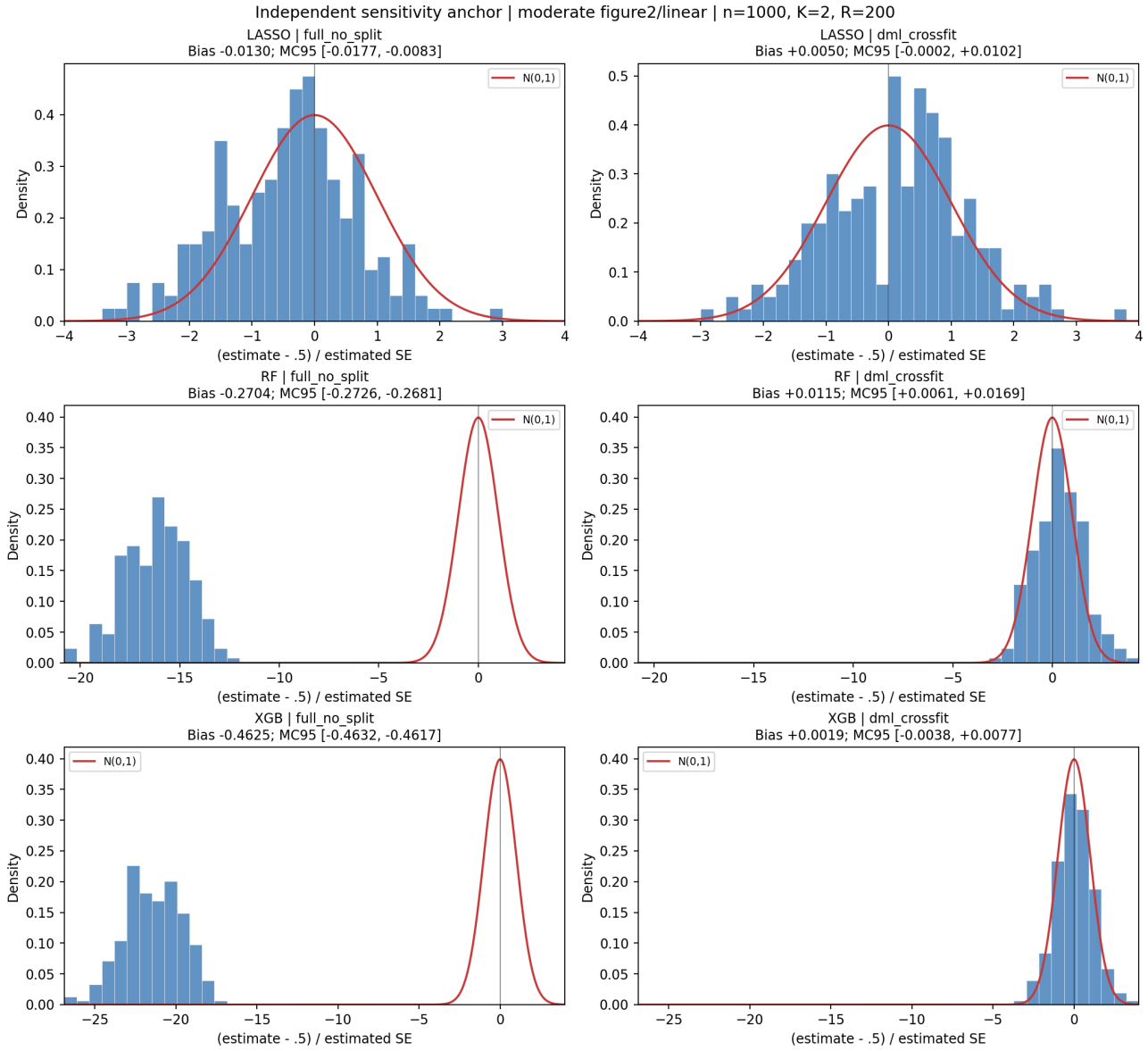

Raw-error companion: full no-split versus cross-fitting (nonlinear moderate DGP, n=1,000, R=200). Every draw is shown on common within-learner axes. This plot is not a studentized-error plot.



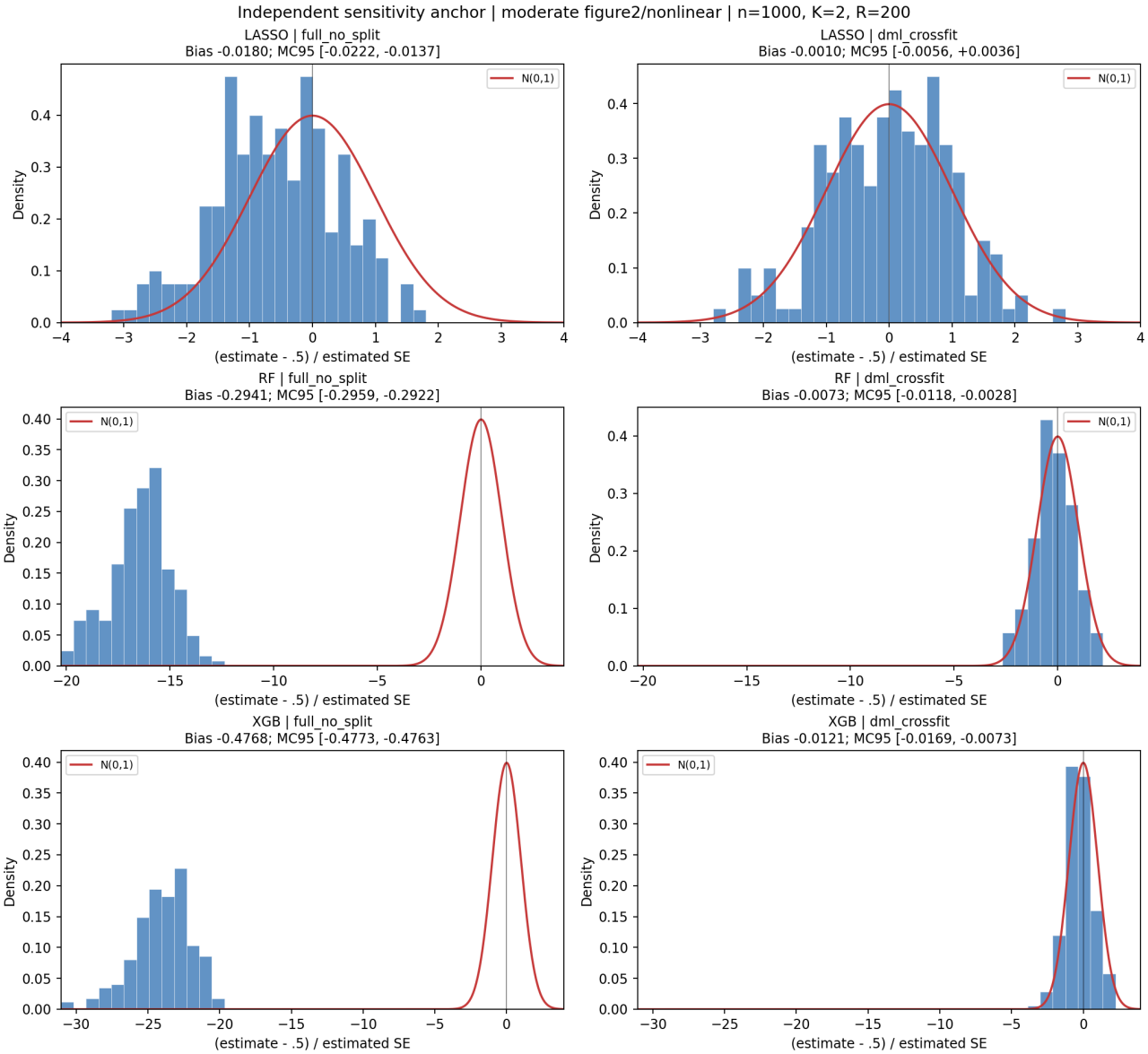

Archived original histogram: figure1 / linear / naive ml. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.



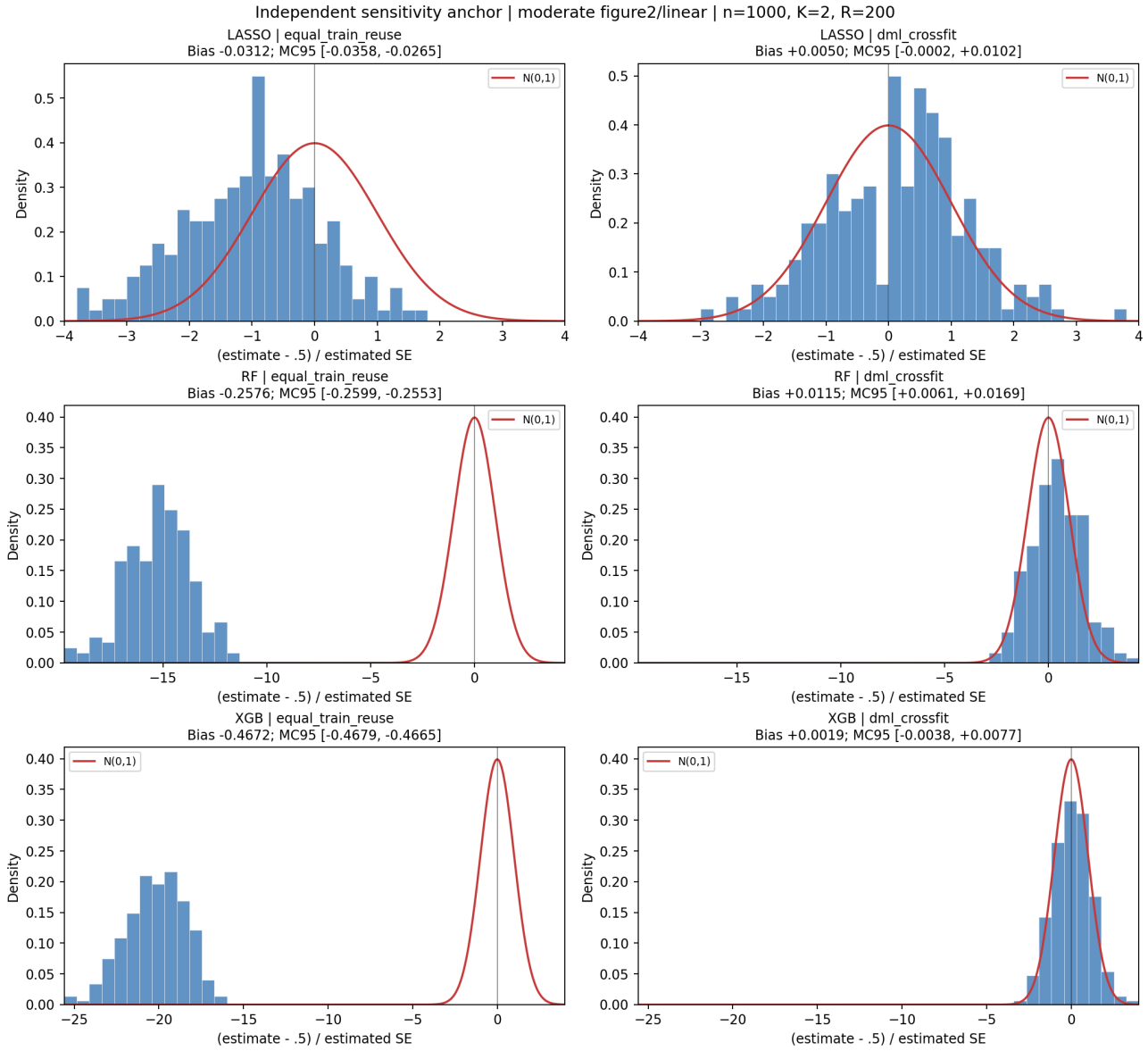

Archived original histogram: figure1 / nonlinear / naive ml. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.



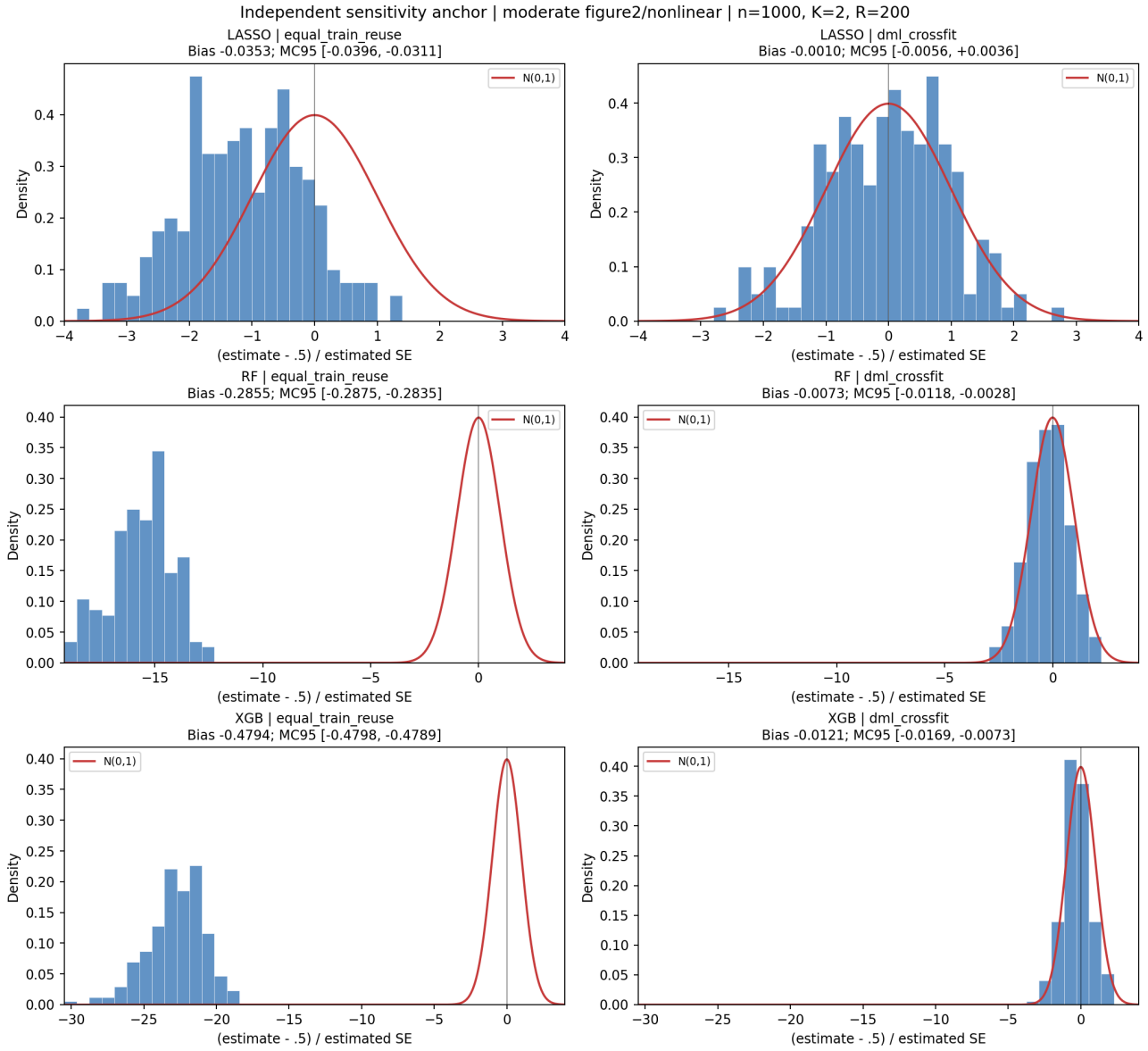

Archived original histogram: figure2 / linear / full no split. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.




Archived original histogram: figure2 / nonlinear / full no split. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.




Archived original histogram: figure2 / linear / equal train reuse. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.




Archived original histogram: figure2 / nonlinear / equal train reuse. Studentized errors; n=1,000, K=2, R=200. This original presentation of the SAME saved experiment is retained for completeness, not counted as a new independent experiment.

## 8. Methods, provenance and claim boundaries

Model: Y=theta0*D+g0(X)+U and D=m0(X)+V, theta0=0.5. X is Gaussian with p=20 and covariance 0.7 to the power |j-k|; U and V are independent standard Gaussian disturbances. Structural g0=E[Y-theta0*D|X] differs from ell0=E[Y|X]=g0+theta0*m0. Only the structural IV-type score (D-m_hat)*(Y-theta*D-g_hat) underlies the headline; partialling-out is not used.

All pilots, scalers and nuisance models use training-fold data only. In the orthogonalization experiment, g is learned from Y-pilot*D; in the reuse experiment, g_hat=ell_hat-pilot*m_hat. Known treatment candidate columns [0,2,4,6] and positive/monotone restrictions are explicit controlled priors, not unrestricted confounder discovery. Exact archived DGP coefficients and moderate additions are retained in the configuration files.

No local fitting, retuning, seed selection or new server run occurred for this attachment. All 800 selected job receipts and 19 frozen source files were hash-checked. The 6,000 selected estimates were recomputed and matched to the saved 600-cell summary to 1e-12. Pairing, prediction overlap, shared nuisance banks, error decomposition and coverage indicators were checked before plotting.

The original raw results, earlier completed simulation chapter and GitHub branch were not modified by this build. Source images are copied unchanged; new figures are deterministic visualizations of the saved records. The complete source summaries retain adverse and inconclusive outcomes, including ordinary-learner residual bias. Histograms are descriptive diagnostics, not proofs of asymptotic normality or universal finite-sample unbiasedness.

Evidence identity: 8386d43620f5ad6b9df45cc88429245d708057e1a7d29ca90cce4d0bba8a8a5a

Machine-readable outputs: summary.json, primary_draws.csv, input_receipt_hashes.json and verification.json. The offline index includes every new table/figure and the complete sensitivity and nuisance summaries.